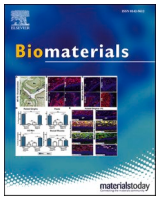
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Skin-interfaced electronics: A promising and intelligent paradigm for personalized healthcare

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ABSTRACT

Skin-interfaced electronics (skintronics) have received considerable attention due to their thinness, skin-like mechanical softness, excellent conformability, and multifunctional integration. Current advancements in skintronics have enabled health monitoring and digital medicine. Particularly, skintronics offer a personalized platform for early-stage disease diagnosis and treatment. In this comprehensive review, we discuss (1) the state-of-the-art skintronic devices, (2) material selections and platform considerations of future skintronics toward intelligent healthcare, (3) device fabrication and system integrations of skintronics, (4) an overview of the skintronic platform for personalized healthcare applications, including biosensing as well as wound healing, sleep monitoring, the assessment of SARS-CoV-2, and the augmented reality-/virtual reality-enhanced human-machine interfaces, and (5) current challenges and future opportunities of skintronics and their potentials in clinical translation and commercialization. The field of skintronics will not only minimize physical and physiological mismatches with the skin but also shift the paradigm in intelligent and personalized healthcare and offer unprecedented promise to revolutionize conventional medical practices.

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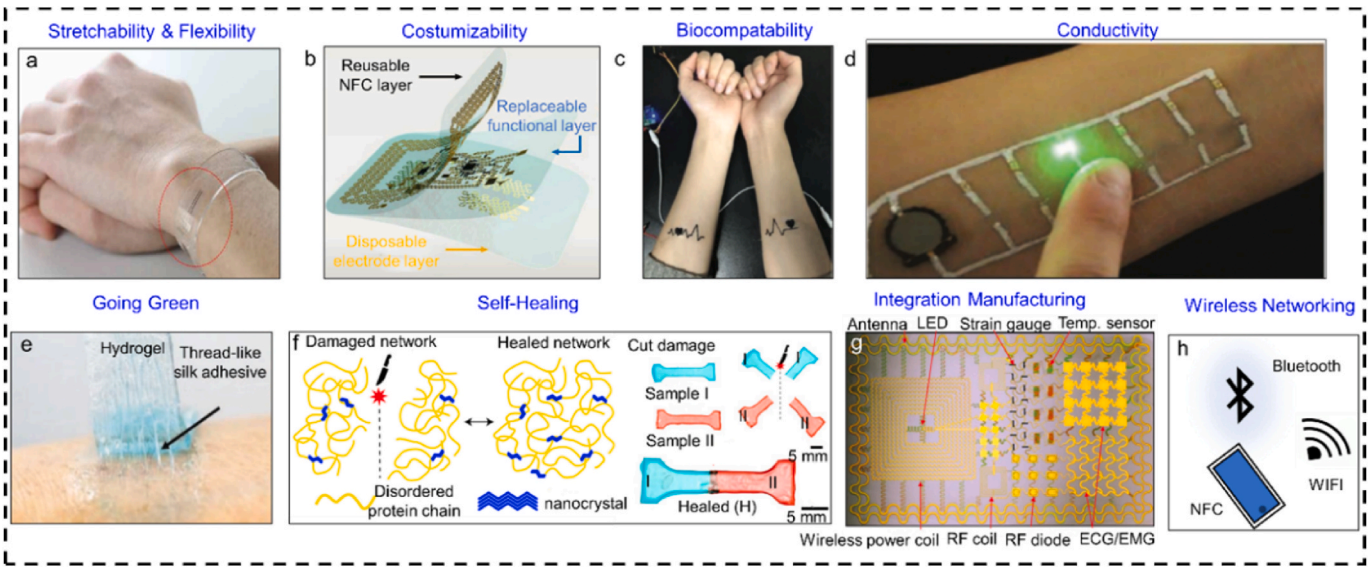


Fig. 1. Overview of the skin-interfaced bioelectronics. The epidermal electronics is expected to have characteristics including stretchability and flexibility (a), customizability (b), biocompatibility (c), conductivity (d), environmentally friendly (e), self-healing (f), integration manufacturing (g), and wireless communication (h). With respect to the material selections. Smart fabrication and integration manufacturing enabled the intelligence of epidermal electronics. Reproduced with permission [321]. Copyright 2019 The authors, American Association for the Advancement of Science; Reproduced with permission [200]. Copyright 2019 Wiley; Reproduced with permission [322]; Copyright 2019 Wiley; Reproduced with permission [94]. Copyright 2019 Wiley; Reproduced with permission [126]. Copyright 2018 Wiley. Reproduced with permission [323]. Copyright 2020, The Author(s), under exclusive license to Springer Nature Limited; Reproduced with permission [21] Copyright 2011 American Association for the Advancement of Science.

1. Introduction

Wearable bioelectronics is a multidisciplinary concept that combines electronics with biological systems. A bioelectronic device can convert biological signals, such as physical, physiological, and metabolic signals, into measurable electrical signals [1–3]. Compared to conventional sampling methods such as blood draws, wearable bioelectronics does not require clinical experts and specialized laboratory equipment, demonstrating the significant potential for health monitoring in a convenient and non/minimally invasive manner [4–7]. The major limitation of commercial wearables, including smart watches, wristbands, and gadgets, is the poor contact with the skin, resulting in lower signal-to-noise ratios. In addition, such wearables are constrained to several specific body positions, significantly weakening the mobility of wearers.

The emergence of skintronic devices can address these concerns by having direct and conformal contact with the skin, even under severe mechanical deformation [8,9]. These devices have been favored to monitor multiple physical, physiological, and biochemical signals emitted from the body without interfering with users' daily routines [10–18]. It is considered an ideal platform for personalized healthcare due to its thinness, softness, and multifunctional integration [19,20].

Since Rogers's group reported the first seminal skintronics example in 2011 [21], research on skintronics are becoming a central scientific area in the past decade, enabling many emerging applications, including health monitoring, fitness or wellness management, disease diagnosis and treatment, and soft robotics. Moreover, signal processing tools such as machine learning (ML) have been employed for data mining, discovering specific disease occurrence patterns, and revealing potential therapeutic opportunities [22–24]. Therefore, skintronics can be evolved into an intelligent platform for point-of-care healthcare in home and community settings.

This review begins with a discussion of material selections and platform considerations for skintronics and then elaborates on device fabrication and system integration (Fig. 1). This review will also highlight real-life healthcare applications, including disease diagnostics and therapeutics. We then envision the market landscape of skintronics from a commercial perspective. Finally, we summarize the current challenges and future directions in the field of skintronics.

2. Material selections and platform considerations for skintronics

Materials and device properties are essential considerations in the design of skintronics. The device in contact with the body must be biocompatible and employ encapsulation or coating techniques to prevent fouling or contamination. Breathability, flexibility, and conductivity should be considered when selecting materials to construct skintronics. Several critical works of literature have provided detailed analyses of available materials for soft electronics [25–27]. In this section, we highlight some of these considerations that innovate the design and development of the skintronic platforms.

2.1. Material selections

Electronic properties play a significant role in constructing skintronic devices. Various conductive materials, which are soft and flexible, have been explored. Herein, we summarize the recent electronic materials, including representative organic polymers and inorganic nanomaterials (*i.e.,* zero dimensional-/one dimensional-/two dimensional-nanomaterials), liquid metals, and ionic liquids.

2.1.1. Zero-dimensional nanomaterials

Various zero-dimensional (0D) nanomaterials, especially gold nanoparticles (AuNPs), have been broadly utilized in biosensing applications due to their facile surface modification and superior chemical stability against oxygen. However, 0D nanomaterials, in the form of nanoparticles, have 10 to 100 times higher percolation threshold than 1D nanomaterials (*e.g.,* nanowires or nanotubes), resulting in insufficient conductivity, which limits their further use in skintronics. To address this issue, AuNPs were assembled into a highly ordered layer by their chemically surface-modification *via* soft capping ligands (*i.e.*, citrate, thiol, and amine) and then were integrated into the stretchable polymer matrices [25]. Kotov et al. developed a stretchable conductor

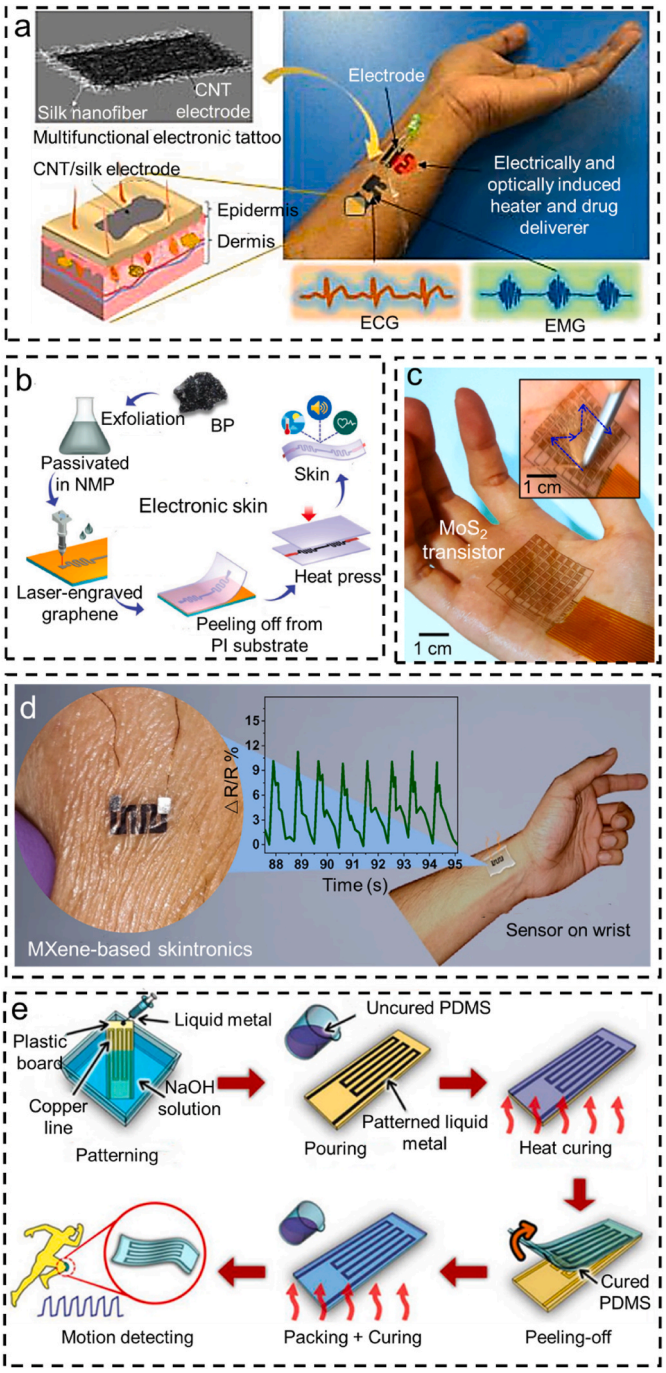


Fig. 2. Representative electrical materials used to construct skin-interfaced bioelectronics. (a) Schematic and optical image of the E-tattoo based on carbon nanotube/silk placed on a human forearm for multiple applications. Reproduced with permission [48] Copyright 2021 Wiley. (b) Schematic illustration of exfoliation of black phosphorus and fabrication of the hybridized electronic skin based on the black phosphorus@laser-engraved graphene. Reproduced with permission [70] Copyright 2020 Wiley. (c) Photograph of the MoS_2-based tactile sensor on a human palm. Reproduced with permission [72] Copyright 2019 American Chemical Society. (d) A MXene-based sensor on the human hand for radial artery pulse monitoring. Reproduced with permission [53] Copyright 2020 American Chemical Society. (e) Schematic illustration showing the fabrication process of the liquid metal-based strain sensor. Reproduced with permission [96] Copyright 2021 Wiley.

with citrate-stabilized AuNPs-polyurethane composite, and it showed excellent stretchability (~480%) and conductivity (~11,000 S/cm) [28]. Alternatively, Cheng and his co-workers reported a monolayered fractal gold nano-framework using a one-pot wet chemistry synthesis method without any capping agent. They directly transferred it onto various flexible substrates (polyethylene terephthalate, polydimethylsiloxane (PDMS), Ecoflex). This gold nanoframework-based conductor could be stretched up to 110% without sacrificing conductivity [29].

2.1.2. One-dimensional nanomaterials

One-dimensional (1D) materials are well-known candidates for developing skintronics due to their tunable conductivity and stretchability [30,31]. Metallic nanowires, especially silver or copper nanowires (AgNWs or CuNWs), have been widely used for skintronics due to their excellent conductivity and flexibility [32,33]. However, the lifetime, reliability, and operating conditions of AgNW-based electronics are challenging because of their poor resistance to oxygen or air [34–37]. Similar issues were also reported for CuNWs [38–40]. Gold nanowires (AuNWs) with excellent biocompatibility and stability have been used for dry electrodes, conductive interconnections between flexible circuits, and skintronic devices [41]. In addition, by tuning the physical layout of AuNWs, a remarkable electronic characteristic can be achieved accordingly [42]. Another form of fractal Au nanomaterials demonstrated some promising features: high conductivity (55.9–100 Ω/square) and high optical transmission (70%–93%) were observed when incorporated into PDMS. Furthermore, a better conductivity and optical transparency system can be obtained with a combinative fractal of Au and Cu [43]. However, the high cost impedes their further commercial applications.

In addition to metallic nanowires, carbon nanotubes (CNTs) have attracted sufficient attention in skintronics because of their superior electrical/thermal conductivity, excellent energy conversion efficiency, mechanical robustness, and chemical stability [44–47]. For example, an ultrathin skintronic named E-tattoo was fabricated by dispersing CNTs onto the silk nanofibrous networks [48]. This E-tattoo could be directly attached to the epidermis and easily removed after use. Multiple functions were demonstrated, including skin temperature monitoring, drug delivery, and electrophysiological signal sensing (Fig. 2a) [48].

2.1.3. Two-dimensional nanomaterials

Two-dimensional (2D) nanostructured materials have been excellent transducers, as they generally exhibit great mechanical flexibility without compromising conductivity [49,50]. Extensive works have been reported with these 2D materials, including MXene, molybdenum disulfide (MoS_2), black phosphorus (BP), metal-organic framework (MOF), and graphene [41,51–56].

Graphene is typical soft conductive material with various advantages such as transparency, robustness, and stability [57–60]. Different graphene-based skintronic devices have been developed [61–63]. For example, Lu's Group reported a graphene-based E-tattoo (GET). The open-mesh structure of the GET provided excellent breathability and negligible mechanical stiffness to ensure the wearer's comfort [60]. Similarly, graphene oxide (GO) and reduced GO (rGO) have also been widely investigated. Oxygen functional groups enable its dispersion into aqueous solutions and functionalization. In addition, GO can be converted to reduced graphene oxide (rGO) *via* physical/chemical approaches [64–66]. Several rGO-based skintronic devices have also been reported [67–69].

Black phosphorus (BP) is gaining attention as a 2D semiconducting material with exceptional features such as a tunable direct bandgap and high carrier mobility. Depending on the structure, the direct bandgap of BPs varies from 0.3 eV for bulk BP to 1.5 eV for monolayer BP. However, using a few-layered BP is challenging due to its fast degeneration under ambient environments and difficulty in surface modification. The passivation of BP flakes with various materials has been investigated to

overcome this issue. Chhetry et al. reported that BP could be combined with CO_2 laser-engraved graphene (BP@LEG) to fabricate a skintronic sensor with high sensitivity (Fig. 2b) [70]. BP nanoparticles were passivated with poly (allylamine) through electrostatic interaction and cast onto the laser-engraved graphene to make BP@LEG composite material, resulting in stable and robust hybridization of the two materials. This BP@LEG electrode deposited into polystyrene-*block*-poly (ethylene-*ran*-butylene)-*block*-polystyrene (SEBS) substrate exhibited a high thermal index (8106 K at 25–50 °C), high strain sensitivity, and durability over 18,400 cycles, demonstrating the feasibility of BP-based electrodes in skintronic devices.

Molybdenum disulfide (MoS_2) synthesized with the chemical vapor deposition (CVD) method shows remarkable semiconducting, optical, and mechanical properties [71]. In addition, MoS_2 has a high gauge factor of 70 for polycrystalline and 200 for single crystal, which is proper for tactile sensing applications. Park et al. reported a large-area, ultrathin, flexible tactile sensor based on MoS_2 with active-matrix (AM) circuitry (Fig. 2c) [72]. MoS_2 was placed between high-*k* Al_2O_3 dielectric layers, and the sensing units of this MoS_2-based tactile sensor were controlled with AM backplane of the thin-film transistor (TFT) array. It enables the multitouch capability, real-time pressure mapping, and precise shape detection by isolating each pressure gauge unit with individual TFT in the AM backplane.

MOFs have great potential in fabricating designable microstructures compared with other conductive materials (*e.g.,* CNTs, and graphene) [73]. In addition, due to its pyro-piezoelectricity coupling effect, MOFs can independently detect temperature and pressure with no interference between each sensing functionality, which makes them good material candidates for dual-modal sensors. But it is challenging to fabricate continuous, stable, and conductive MOF films with desired microstructures because of the low conductivity of most MOFs [74]. Besides, conductive MOF films suffer from low resistance against tensile strain. To address this issue, Li et al. recently reported a pressure-temperature dual-modal sensor fabricated with a micro-structured MOF film which is highly conductive, flexible, and stain-resistant [75]. This MOF film was manufactured by *in situ* growing $Ni_3(HiTP)_2$ onto micro-structured mixed cellulose (MSMC) substrates, achieving a continuous and highly conductive (4000–5000 S/m) microstructured MOF. With higher tensile Young's modulus than PDMS, the resistance and thermal current of MOF-MSMC composite films were not affected by stretching, making it possible to accurately measure pressure and temperature under high tensile stress. Consequently, this MOF-MSMC-based sensor demonstrated excellent performance in pressure-sensing and temperature-sensing, which are superior to conventional dual-modal sensors. In another work, the moisture permeability of MOF is further harnessed by combining with nano-cellulose to construct a full sweat-proof sensor system skintronics [76].

MXene, a new family of 2D materials, has been used to engineer skintronics because it exhibits excellent conductivity, extraordinary mechanical properties, and hierarchical structures [53,77]. MXene-based applications in electrodes, sensors, antennas, and supercapacitors have been exemplified in various ways [78–81]. For skintronics, persistent efforts have been focused on skin-conformal capability using MXene [78,82,83]. Kedambaimoole et al. adopted laser-induced direct patterning to fabricate freestanding films for skin sensors. The high sensitivity with a gauge factor of 7400 was achieved due to the unique nanostructures induced by MXene (Fig. 2d) [53].

In addition, 2D transition metal dichalcogenides (TMDCs) with a generic MX_2 composition (M = Mo, W, Pt, *etc.*; X = S, Se, Te) have been widely studied for their physical properties and electronic applications [52,84]. Besides semiconductive/metallic properties, ultrathin (<1 nm), soft, and flexible features made TMDCs suited for epidermal electronics [85]. For example, platinum diselenide and platinum ditelluride layered materials have been used in skintronic devices to obtain physical and biopotential information, such as temperature, electroencephalography (EEG), and electrocardiography (ECG) [52].

2.1.4. Liquid metal

Liquid metal (LM) materials refer to metal/metal alloys that are liquid at/near room temperature [86]. They have been used in wearables and implantable bioelectronics due to their flexibility and high conductivity. Among the LM materials, gallium (Ga)-based alloys have been considered attractive owing to their excellent electrical/thermal conductivity, mechanical transformability, self-healing, and non/low toxicity [87–89]. However, compared to other LMs, the Ga-based alloys exhibit high surface tension (>600 mN/m) and self-limiting surface oxidation under ambient conditions and tend to form a spherical shape which makes them difficult to process [90–92]. LM patterning techniques, including laser patterning, micro-contact patterning, and three-dimensional (3D) printing, have been adopted to address these issues.[93] For example, eutectic gallium-indium (EGaIn) and Ga-based LM materials were used with a methacrylate glue to fabricate E-tattoos [94]. With functionally integrated units, the E-tattoos could be used for ECG and electromyography (EMG) detection and temperature monitoring, acting as a human-machine interface to recognize body movements and gestures. This work presented a facile and low-cost approach to LM-based skintronics. However, the concerns about LM materials' compatibility and elasticity remain to be addressed [95]. Wu et al. adopted a selective wetting and transfer process to improve the compatibility between Ga-based LM materials and PDMS substrates for constructing a stretchable resistive strain sensor. This device could detect a broad range of strains with a low detection limit, negligible hysteresis, superior mechanical robustness, short response time, and good repeatability (Fig. 2e) [96].

2.1.5. Ionic liquid

Ionic liquids (ILs) have been increasingly gaining attention since the first report of room-temperature molten salt ethylammonium nitrate ($[C_2H_5NH_3][NO_3]$) by Walden in 1914 [97]. Recent advances in synthetic chemistry allow ILs to be chemically modified for skintronics due to their intrinsic flexibility, adhesiveness, and no/low toxicity [98–101]. Chen and colleagues developed a dynamic ionic liquid (DIL) for manufacturing skintronics. The DIL units were polymerized into linear polymers and crosslinked to form ionic gels [100]. This skintronic device was conformably attached to human skin without irritation. The superior conductivity of ILs enabled excellent performance in sensing temperature and tensile strain changes, demonstrating the potential of ILs in skintronics [100].

2.2. Platform considerations

To promote long-term use, a skintronic platform should be ultrathin, skin-biocompatible, and conformable. Concerning sustainability and environmental safety, degradable materials with minimal disposal contamination are desired. Hence, this section provided a comprehensive summary of critical considerations for constructing skintronic platforms for personalized healthcare.

2.2.1. Flexibility, stretchability, and skin-conformability

Flexibility and stretchability are essential to provide robust and stable human health monitoring, especially during strenuous activities such as cycling and running [20,102]. Strategies from macroscopic to molecular-scale designs have been proposed based on the following three principles: (1) the use of intrinsically-elastic materials, (2) hybrid methods to achieve stretchability, and (3) specific pattern designs with excellent stress tolerance. Intrinsically stretchable materials can be obtained after proper chemical modifications, as noncovalent chain-chain interactions among such modified polymers can facilitate dynamic bonding and energy dissipation [103]. Based on the percolation theory, conductive fillers within elastomers have been used to develop highly stretchable skintronics [104,105]. In this respect, hybrid elastomer composites integrated with nanomaterials, such as NPs and NWs, have been utilized for skintronics design [106–112]. Serpentine structures

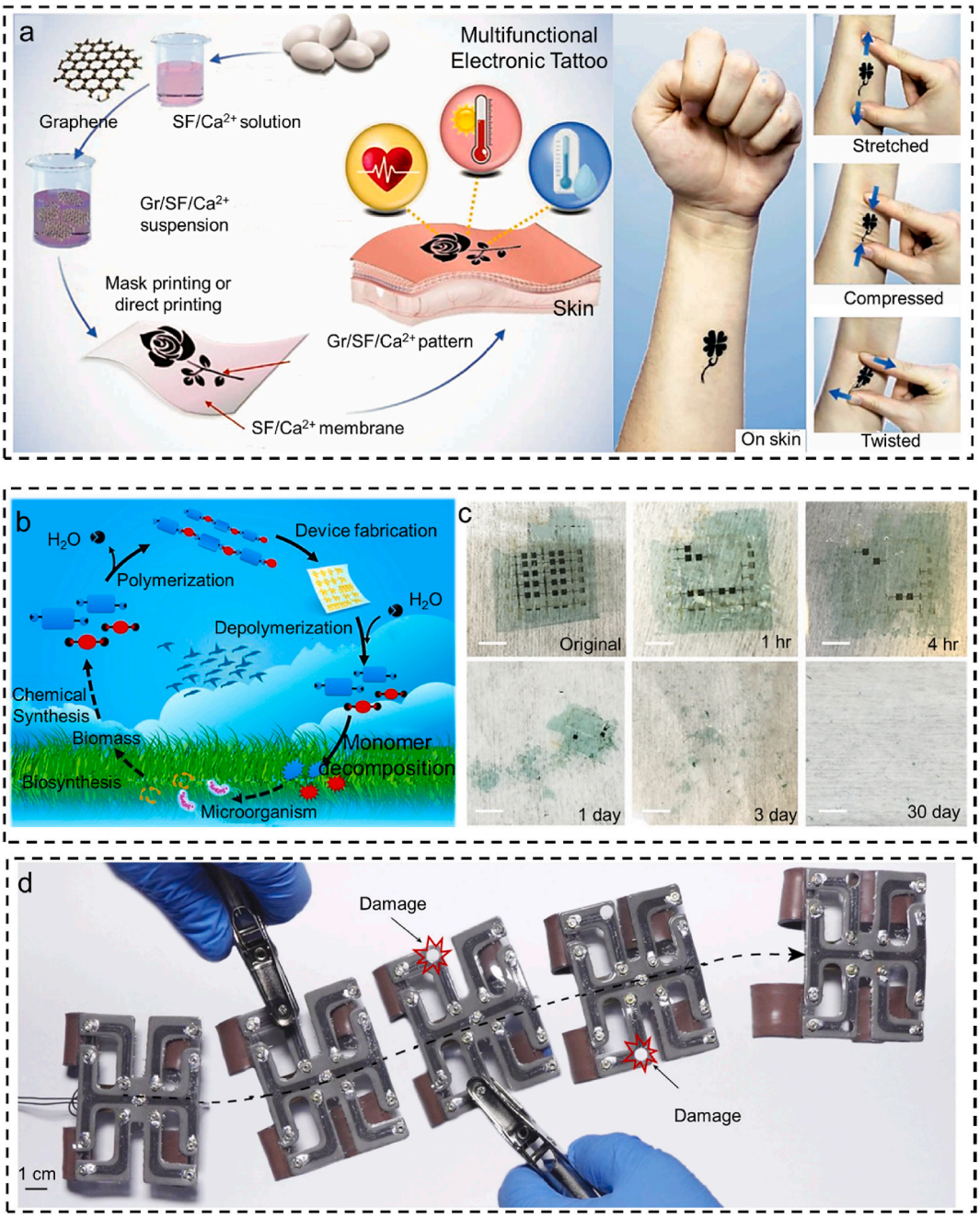


Fig. 3. Platform considerations for skin-interfaced bioelectronics. (a) Schematic illustration of the graphene/silk/Ca^{2+} tattoo with its compatibility-stretchability-conductivity balance. Reproduced with permission [322] Copyright 2019 Wiley. (b–c) Schematics of the transient green skintronics and its degradability-conductivity balance. Reproduced with permission [143] Copyright 2017 PNAS. (d). The electrical pathway of the soft robot is autonomously reconfigurable around the damaged region without loss of electrical conductivity once the robot encounters puncture damage, indicating excellent self-healing capability. Reproduced with permission [161] Copyright 2018 Nature Group. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

can benefit skintronics design by conductive interconnectors of rigid electronic components on elastic polymeric substrates. Such architectural design enables a possible biaxial strain tolerance, indicating excellent skin-conformability [113,114].

2.2.2. *Biocompatibility-conductivity balance*

Skin compatibility is essential for skintronic devices to avoid potential skin irritation [115]. However, an optimal skin compatibility-conductivity balance is challenging as most biocompatible substrates are insulating polymers, such as gelatin methacryloyl (GelMA) and polyvinyl alcohol (PVA), and silk fibroin (SF) [116–122]. To address this issue, conductive materials have been integrated to realize conductive polymeric elastomers [123–125]. For example, Choi and colleagues developed a biocompatible and conductive nanocomposite by mixing Au-coated Ag nanowires into poly (styrene-butadiene-styrene) (SBS) elastomers. The Au coating could minimize Ag oxidation, allowing it to retain a high conductivity (approximately 41,850 S/cm^2) without sacrificing the stretchability of SBS substrates (around 266%) [123]. Another example by Seo et al. reported a skintronic device based on calcium (Ca^{2+})-modified SF bioadhesive [126]. The introduction of Ca^{2+} facilitated the viscoelastic property and mechanical interlocking of the SF network [127]. Zhang's

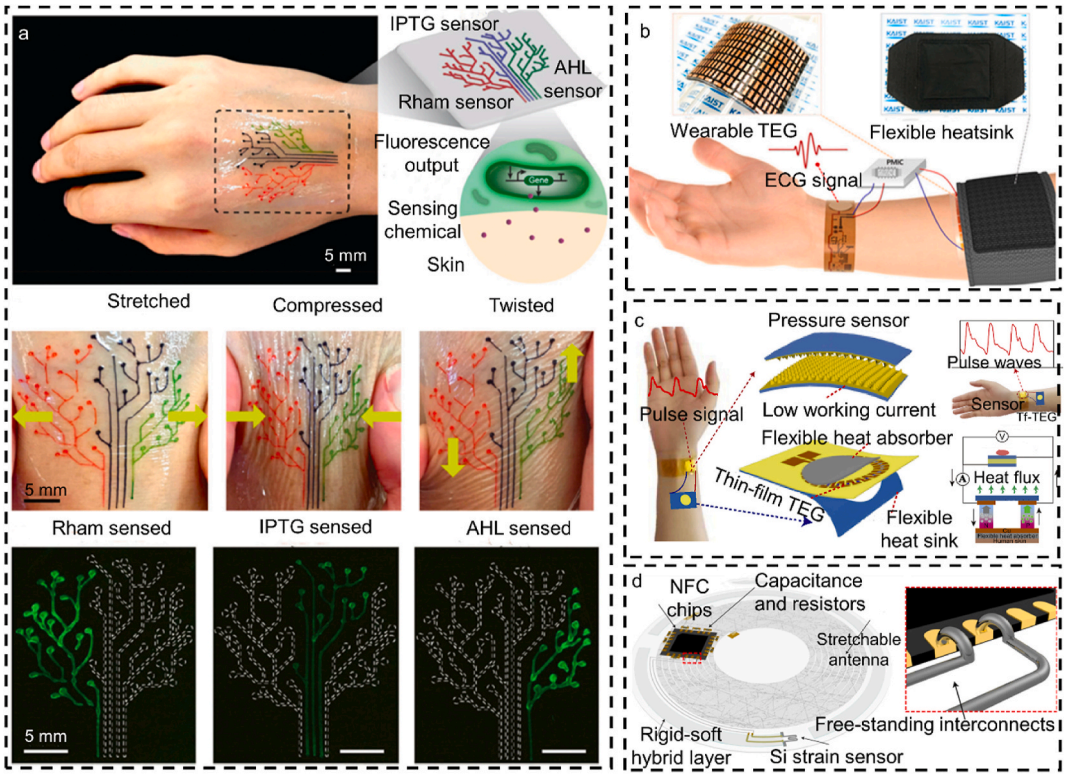


Fig. 4. Examples of skin-interfaced bioelectronics employed for various biomedical applications. (a) 3D-printed living tattoo for chemical detection on human skin. Top: 3D-printed living tattoo platform with the detection ability for chemicals for its application to the skin surface. Middle: The illustration of the tree-like pattern on the thin elastomer layer directly applied to the skin surface. Different colors refer to the differences in cell types, which can react to various chemicals and emit corresponding fluorescence. The living tattoo showed excellent tolerance for stretching, compression, and twisting with great mechanical characteristics. Bottom: The demonstration of the living tattoo to different chemicals: Rham, IPTG, or AHL. Reproduced with permission [185] Copyright 2017 Wiley. (b) Schematic illustration showing the self-powered wearable ECG system. Reproduced with permission [190] Copyright 2018 American Chemical Society. (c) Schematic description of the self-powered pressure sensor based on a flexible thin-film thermoelectric generator and the equivalent circuit diagram. Reproduced with permission [191] Copyright 2020 Elsevier. (d) Schematic of a fully integrated wireless system, including stretchable silver nanofiber-silver nanowire antenna, NFC chip, passive components, and stretchable liquid metal conductive interconnections. Reproduced with permission [324] Copyright 2021 Nature Group. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

Group employed silk/graphene-based conductive inks to construct skintronics that can achieve good biocompatibility and conductivity (Fig. 3a) [128]. In addition to employing biocompatible materials, improving the breathability of the skintronics by employing nano/open mesh structures is another strategy to enhance skin compatibility and avoid hypoallergenic reactions [129–132]. In this regard, PVA-based conductors with nanomesh structures displayed excellent gas permeability for skintronic devices [131–133].

2.2.3. *Biodegradability-conductivity balance*

Biodegradable materials are desirable candidates for temporary electronics, particularly disposable bioelectronics [134–138]. These biodegradable materials can be synthetic or naturally derived [139]. Specifically, commonly-used synthetic biodegradable polymers include polyvinyl alcohol (PVA), polycaprolactone (PCL), and poly (lactic-*co*-glycolic acid) (PLGA), while typical natural polymers include cyclodextrin, cellulose, chitosan, hyaluronan, and collagen. Degradable and resorbable biomaterials could eventually help to fabricate eco-friendly skintronics [115,140–142].

However, these biodegradable polymers used to make skintronics suffer from poor conductivity. To address the issue, Bao's group reported a biocompatible and fully disintegrable semiconducting polymer with chemically modified diketopyrrolopyrrole (DPP) that could be decomposed under mildly acidic conditions [143]. The ultrathin film was designed as a biodegradable substrate, combined with lightweight pseudo-complementary metal-oxide-semiconductor (CMOS) logic circuits, to build green organic skintronics with biodegradable capabilities (Fig. 3b–c) [143]. Similarly, Bettinger et al. fabricated a degradable electronic transistor by depositing a small-molecule semiconductor-5, 5′-bis-(7-dodecyl-9H-fluoren-2-yl)-2,2′-bithiophene (DDFTTF) on the resorbable PLGA substrate. They demonstrated that this transistor retained its functionality after exposure to water and was also resorbable under biological conditions [144].

2.2.4. *Self-healing*

In nature, self-healing is a crucial feature at the molecular level (*e.g.*, DNA self-repair) and the macroscopic level (*e.g.*, wound closure) [145]. Likewise, the self-healing property has gained much attention in material design since severe mechanical stress can lead to fatal damage to materials and their functionalities [146]. Therefore, skintronics with self-healing ability have become a new option to ensure a long service period [147,148]. Although several self-healing polymers have been used in practice, it remains challenging to recover the mechanical and electrical properties after damage completely [147–157]. Typically, self-healing originates from physically and/or chemically dynamic and reversible intermolecular interactions [158,159]. The use of soft elastomeric composites and biopolymers has been suggested for skintronics [126,160]. Majidi's group reported a novel architectural design for soft and deformable skintronics. The extraordinary resilience to mechanical damage was realized by suspending liquid metal droplets (LM) into the soft elastomer, as shown in Fig. 3d [161].

2.2.5. *Anti-bacterial*

Bacterial infections are another concern for the practical applications

of skintronics due to its usage conditions with prolonged exposure to air or moisture and various climate environments [162–167]. To develop skintronics with antibacterial performance, Zhu and colleagues incorporated the antibacterial agent-glycerol monolaurate compounds into the PDMS substrate [168]. Some other reports employed AgNW film or zinc oxide (ZnO) nanostructures as anti-bacterial components of the skintronics [169–171].

3. Device fabrication and system integration

3.1. Device fabrication

High-quality skintronic devices depend not only on the appropriate materials but also on fabrication techniques. Many fabrication techniques, such as transfer, screen, stencil, and 3D printing, have been developed [172,173]. However, constructing skintronic devices for scalable manufacturing and applications remains challenging. This section overviews device fabrication and system integration for personalized healthcare.

3.1.1. Transfer printing

Conventional electrode fabrication requires several procedures to manufacture electrodes with sacrificial layers and tattoo papers and transfer them into skintronics [174,175]. Although acoustic/pressure-based transfer techniques have been established, it still requires specific temperature (90–120 °C) and applied pressures (up to 104 psi), which limits their practical applications [176–178]. Tattoo-like transfer strategies are developed to address these issues. For example, Piva and colleagues used a transfer technique by printing poly (3,4-ethylene dioxythiophene) polystyrene sulfonate (PEDOT: PSS) on a tattoo paper, followed by the direct transfer onto organic photoactive layers. After the starch-dextrin-based tattoo paper dissolution, top PEDOT: PSS contacts were assembled [174]. This strategy achieved ease of operation and low power supplies, paving the way for scalable and high throughput manufacturing.

Later, Fang et al. reported a cryo-transfer method for constructing ultrathin, stretchable, and skin-conformable electrodes composed of AgNWs and elastomeric polymers [175]. This method allowed the soft elastomers to temporarily maintain a higher elastic state when immersed in liquid nitrogen, and the generated gripping force promoted the electrode transfer. The AgNWs-elastomer composite could rapidly recover to its room-temperature state without material damage and conductivity changes. Compared to elastomer stamp/sacrificial layer strategies, the transfer method was easier to operate. It was applied to low-elastic substrates such as hydrogels, thereby facilitating the integration of printable electronics with soft, skin-like substrates to achieve more comfort for human use [102,111].

3.1.2. Additive manufacturing for skintronics

Additive manufacturing, also known as 3D printing, has been widely used in drug delivery, tissue regeneration, stimuli-responsive materials, and shape-memory biomaterials [179–184]. Zhao's group engineered a living tattoo skintronic device by combining 3D printing and genetically programmed cells [185]. The skintronics was designed as a bilayered elastomeric sheet with chemical-sensible cells. This living tattoo could be attached to human skin and showed excellent deformability under various movements. Three signaling chemicals were separately applied to the skin, and green fluorescence could be noticed in different signal-detection areas (Fig. 4a). This material design represents a crucial advance in harnessing living components for skintronics. The 3D printing method was critical in fabricating the microstructures with a designed distribution of cells and chemicals within the hydrogel matrices. Therefore, it is required to elaborate further on the 3D printing method for printing high-resolution living materials that can integrate programmed cells into skintronic devices maintaining high cell viability and functionalities such as sensing and on-demand therapeutic production [185].

3.2. System integration

3.2.1. Wearable powering

Batteries have been the primary energy source for portable electronics for decades; however, their rigidity, weight, size, toxicity, and intermittent charging or replacement do not comply with the requirements of skintronics [186]. Hence, alternative reliable energy sources such as chemical, kinetic, and thermal have been researched to implement self-powered epidermal devices [187,188]. Wearable devices strive to harvest chemical energies through body fluids such as tears and sweat; nevertheless, the energy density of such sources is insufficient to power up most electronics and sensors. The kinetic energy from human motion is a renewable source capable of providing large energy densities; however, due to its sporadic nature, the applications are limited to athletic devices. Heat energy, on the other hand, has perceived tremendous enthusiasm for usage in wearable devices and will be discussed here. The heat dissipated through the skin is a sustainable, renewable energy source that can be transduced to electrical energy.

In a thermoelectric generator (TEG), a temperature difference across an electrically conductive material, also known as a thermoelectric leg, causes migrations of the valence electrons from the hot to the cold side. Hence, an electric field (voltage) is induced across the material. In addition to metals, inorganic semiconductor thin films such as p-type bismuth telluride (n-Bi_2Te_3) and n-type antimony telluride (p-Sb_2Te_3) are the best thermoelectric legs at near room temperatures [189]. Since the temperature difference between the human body and its ambient is relatively low, the amount of power generated by a TEG is insignificant. Therefore, a wearable TEG utilized a heat sink on the ambient side to increase the temperature difference and power [190]. The device generated 8 μW/cm^2 at a steady state by using a thin, flexible metallic heat sink. The device generated 38 μW/cm2 within the first 10-min operation by replacing the metal with a flexible polymer. Gradual evaporation of the polymer's water content resulted in less power generation; nevertheless, it was 65% higher than the case of the metallic heat sink, even after 22 h of continuous operation. The performance of a fabricated TEG, with a surface area of 40 cm^2, was validated by driving a wearable ECG (Fig. 4b).

To further improve the performance, another study used a flexible composite film of PDMS and boron nitride (BN) as a heat absorber on the skin side [191]. The device generated an open circuit voltage (V_{oc}) of 8.2 mV without a heat sink. Including a hydrogel heat sink with 80% water content increased the V_{oc} to 15.8 mV (Fig. 4c).

3.2.2. Wireless communication

Electromagnetic waves have revolutionized the transmission of information over distances without requiring physical media, relying on a wide range of frequencies, including radio frequency (RF), microwaves, and infrared (IR) [187]. Wireless fidelity (Wi-Fi), Bluetooth (BT), ZigBee, and near-field communication (NFC) are among the standardized protocols defining communication rules, frequency, range, and speed. BT and NFC provide low-power solutions for communications between sensors and computing platforms (*e.g.*, microcontrollers and processors); however, BT provides a more extended range of about 10 m using a frequency of 2.4 GHz.

NFC uses electromagnetic induction between two conductive coils in direct proximity of a few centimeters. It can be used for electrical power transfer to low-power electronics (~10 mW) and data communication using a frequency of 13.56 MHz and a speed of up to 424 kbit/s [192]. Using the NFC protocol, a wearable soft contact lens has demonstrated wireless transmission of real-time intraocular pressure (IOP) to monitor glaucoma progression without a battery [193]. The wireless system included a stretchable transparent antenna, resistors, capacitors, and a commercial NFC chip. The antenna was fabricated by a random network of ultralong silver nanofibers (AgNFs) and AgNWs. A human pilot trial

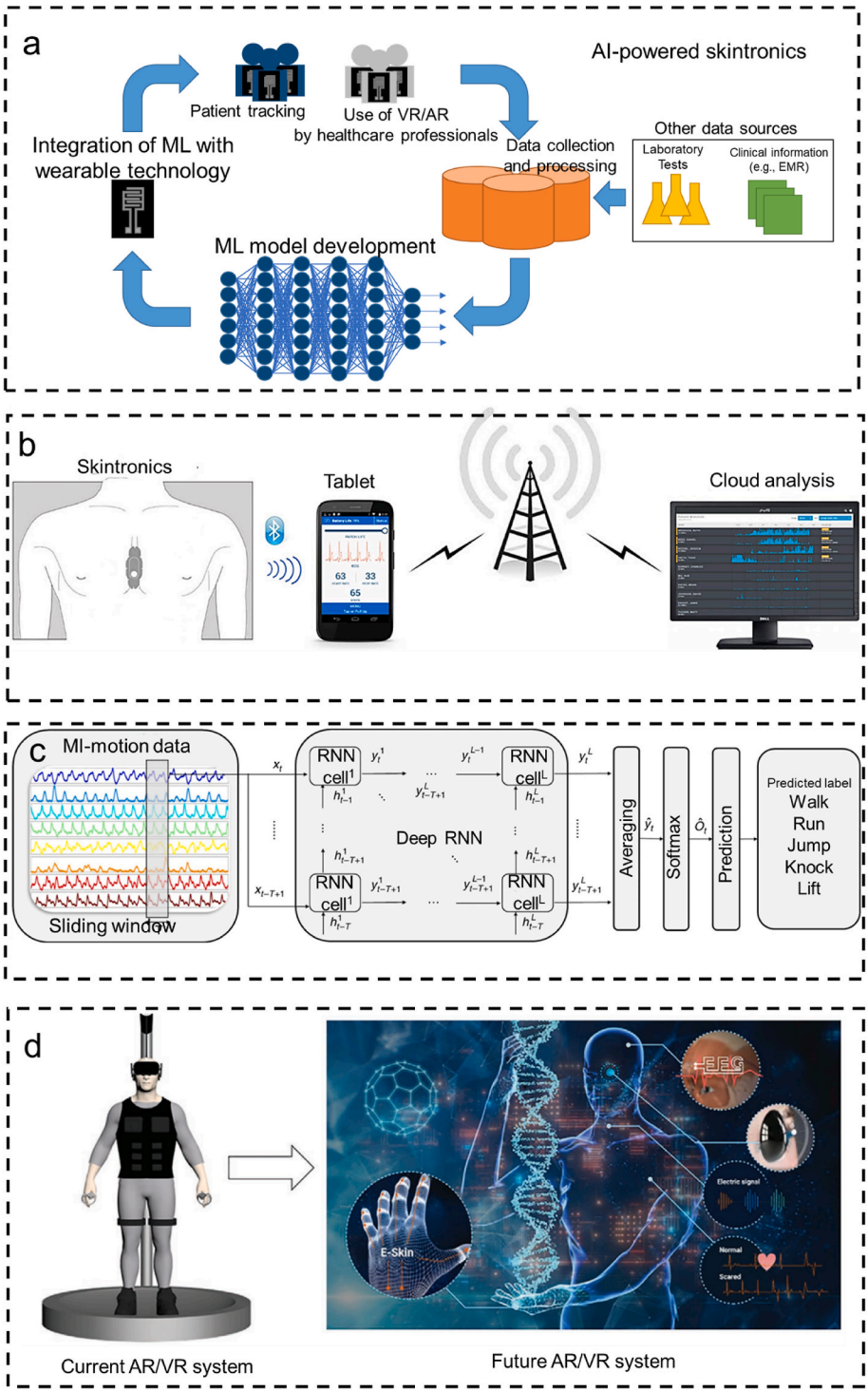


Fig. 5. Artificially intelligent-powered skin-interfaced bioelectronics. (a) Schematic diagram showing a potential implementation cycle to integrate machine learning for the next-generation skintronics. (b) Data collected by the skintronics are streamed to a phone and then encrypted and uploaded to a cloud analytics platform. Reproduced with permission [224] Copyright 2020 American Heart Association. (c) The architecture of deep recurrent neural network for human activity recognition. Reproduced with permission [230] Copyright 2020 Nature Group. (d) The concept of future skintronics for AR/VR systems. Reproduced with permission [325] Copyright 2021 Wiley.

and *in vivo* studies on live rabbits showed good thermal and electromagnetic radiation compatibilities (Fig. 4d).

Another challenge for wireless skintronics is impedance matching at high strain levels. The antenna occupies a relatively large surface area in a wireless subsystem compared to the passive and active auxiliary components. Thus, dynamic strains result in variations of the inductance L, parasitic capacitance C, and series resistance R_S of the antenna, distorting its transmission performance. Based on magnetic resonant coupling, the effects of the antenna's shape and strain on its lumped parameters have been studied to fabricate a transparent wireless stretchable system for power delivery and data transmission [194]. A spiral coil and electronics were fabricated by photolithography and wet etching of AgNFs on an elastomeric PDMS. The results showed that geometry with an infinite number of edges (*i.e.*, a circle) is more appropriate for the shape of the antenna. It was also demonstrated that the inductance of the coil remains dominant at low tensile strains, whereas an increase in the strain induces more cracks which inevitably increases C and R_S. In the presence of higher strains, the cracks become larger, causing the capacitive effect to dominant. Therefore, depending on the required specifications, a well-designed epidermal antenna and circuitry may balance stretchability and transmission efficiency.

3.2.3. Configurable skintronics design

Customizable design is based on using several single-functional building blocks to fabricate multifunctional bioelectronic devices. Multifunctional skintronics have been constructed for various biosensing applications, such as blood glucose levels, skin temperature, and sweat analysis [195–198]. However, most integrated skintronic devices are designed for specific sensing tasks and will be discarded after a single use, which can cause a considerable waste of the circuits [199]. Customizable design strategies are leveraged to address this issue. For example, Lu's Group offered a modular and customizable E-tattoo platform in a multilayered fashion [200]. The wireless E-tattoo platform contains three functional layers: a near-field communication (NFC) layer, a circuit layer, and a passive electrode/sensor layer. The manufacturing process facilitated the selection and easy exchange of proper sensing modalities for E-tattoos as a personalized healthcare platform [200]. Another type of customizable skintronics was implemented *via* the *in situ* and operando adaptation (SOA) method [201]. This fabrication enabled evolving modification and on-demand functional incorporation for skintronic devices. The immediate SOA construction method of the skin electronics provided concurrent customization of skin sensors and wireless functions for personalized healthcare.

3.2.4. Artificial intelligent skintronics for personalized healthcare

Recent years have seen significant advances in artificial intelligence (AI) and computational power that have bolstered the implementations of AI by industries. In particular, there has been a surge of interest by the healthcare sector in using AI, particularly machine learning (ML) and its subfield, deep learning (DL), to aid clinicians with various medical tasks such as patient monitoring, disease diagnostics, treatment planning, and clinical decision-making [202]. While certain ethical and technical challenges remain in implementing AI in the clinics due to concerns surrounding model bias, interpretability, and generalizability [203,204], AI has rendered it possible to effectively and collectively use data on individual patients to promote the idea of personalized medicine further. Particularly, there has been increasing attention to the use of DL. DL requires less manual engineering of the input features in a given dataset for model development, which reduces the loss of information during the data pre-processing. However, it comes at the cost of diminished model interpretability [205]. For example, DL models such as convolutional neural networks (CNNs) have shown tremendous potential in image-based diagnostics using modalities such as MRI and X-ray with accuracy comparable, if not superior, to human experts [206,207]. In addition to medical images, to achieve the aforementioned clinical tasks, DL can employ various sources of data, including but not limited to: (1) electric medical records [208–210]; (2) omics data [211,212]; (3) laboratory data [213,214], (4) medical claims [215,216], and; (5) wearable sensor data [217–219].

Skintronics is one emerging field that has benefited from the development of AI, which can augment the utility of such a platform [220,221]. The outstanding capability of skintronics to capture patient data efficiently and reliably in a real-time, non-invasive fashion makes it an ideal device for collecting large amounts of high-quality data, which is essential for developing robust ML models (Fig. 5a). As such, substantial efforts have been made to integrate wearable sensor technology and AI for various medical applications such as preventative medicine, disease diagnostics, and patient monitoring. For example, a recent study demonstrated that smartwatches coupled with ML algorithms might be used to predict clinical laboratory measurements such as hematologic tests [222]. The authors first processed data collected on heart rate, skin temperature, electrodermal activity, and the number of steps and engineered digital features based on descriptive statistics and expert opinions. The selected features were subsequently used to develop three regression models: random forest, least absolute shrinkage selection operator (LASSO), and canonical correlation analysis (CCA). The random forest model significantly outperformed LASSO and CCA and, of the 44 clinical tests examined, predicted four hematologic tests (*i.e.*, hematocrit, red blood cell count, hemoglobin, and platelet count) with the highest correlation.

The use of wearable sensors coupled with ML also holds promise in enhancing the management of patients with cardiac disorders such as heart failure (HF) [218,223]. Recently, a clinical study also known as the LINK-HF study explored the use of a wearable electronic sensor to collect physiological data such as ECG signals, 3-axis accelerometer, skin temperature, physical activity, and body posture to predict the risk of re-hospitalization after initial HF admission (Fig. 5b) [224]. Collected data were then transmitted to a master device (*i.e.,* an Android smartphone), encrypted, and uploaded to a cloud-based analytics platform to develop a personalized computational model. A multivariate change score was calculated from vital signs *via* similarity-based modeling to quantify the magnitude of change in patients' status compared with their baselines. When looking at a time window of 10 days before hospitalization for performance evaluation, the algorithm achieved 76% sensitivity and 84.8% specificity for HF re-hospitalization.

Epilepsy or seizure is another common, devastating health problem in which patient monitoring *via* AI-powered wearable sensors may be critical for effective disease management [225,226]. In a recent study by Tang et al. [227], a wearable sensor worn on the wrist or ankle was used to measure body temperature, electrodermal activity, accelerometry (ACC), and blood volume pulse (BVP). It was employed to (1) classify different types of seizures, such as generalized tonic-clonic seizure (GTCS), and focal to bilateral tonic-clonic seizure (FBTCS); or (2) detect any seizure episode. Collected data were pre-processed with a sampling technique to adjust the class balance and used to train and evaluate CNN models. The model trained with a fusion of ACC and BVP data yielded the highest leave-one-out area under the receiver operating characteristic curves (ROC-AUCs) of 0.976 and 0.932 for the detection of GTCS and FBTCS, respectively. Similarly, the general model for detecting any seizure type achieved the highest cross-validated ROC-AUC of 0.752 when a fusion of ACC and BVP was used for model development.

Human activity recognition has been one of the most studied topics within the field of intelligent sensors owing to its potential to improve quality of life in various medical areas such as sports medicine, remote patient monitoring for people with motor or cognitive disorders, senior care, and rehabilitation [228,229]. In this regard, Golestani et al. introduced the use of an energy-efficient and cost-effective smart magnetic induction-based HAR (MI-HAR) system to monitor physical movements by magnetic induction (MI) signals, wireless transmission, and deep recurrent neural networks (DRNNs) (Fig. 5c) [230]. Synthetic MI motion data was used to train various ML algorithms, including long short-term memory DRNN (LSTM-DRNN), support vector machines, K-nearest neighbors, decision trees, random forests, and logistic regression. As a result, the LSTM-DRNN model, with overall accuracies of 87.0 and 98.9 for two datasets, outperformed the rest of the algorithms in classifying various physical movements such as knocking, lifting, and walking.

Another interesting application of wearable sensors and AI is for human-machine interfaces to promote the usage of virtual reality/augmented reality (VR/AR) by healthcare professionals to improve the quality of patient care (Fig. 5d) [231–233]. Some significant use cases include providing healthcare personnel in highly technical areas, such as surgeons and radiologists, with effective education and training, enhancing visual fields during a surgical operation, and helping with pre-operative planning using anatomical images in VR/AR. As traditional VR/AR devices (*i.e.,* head-mounted displays) may present certain limitations, such as user discomfort and data latency, the development of more advanced, light-weighted, wearable electronics specialized for VR/AR applications will improve user experience and facilitate the implementation of VR/AR technology within clinical settings. ML algorithms may be developed using various data sources such as pre-operative anatomical images, clinical information, and wearable sensors to promote its usage and further integrate into the VR/AR

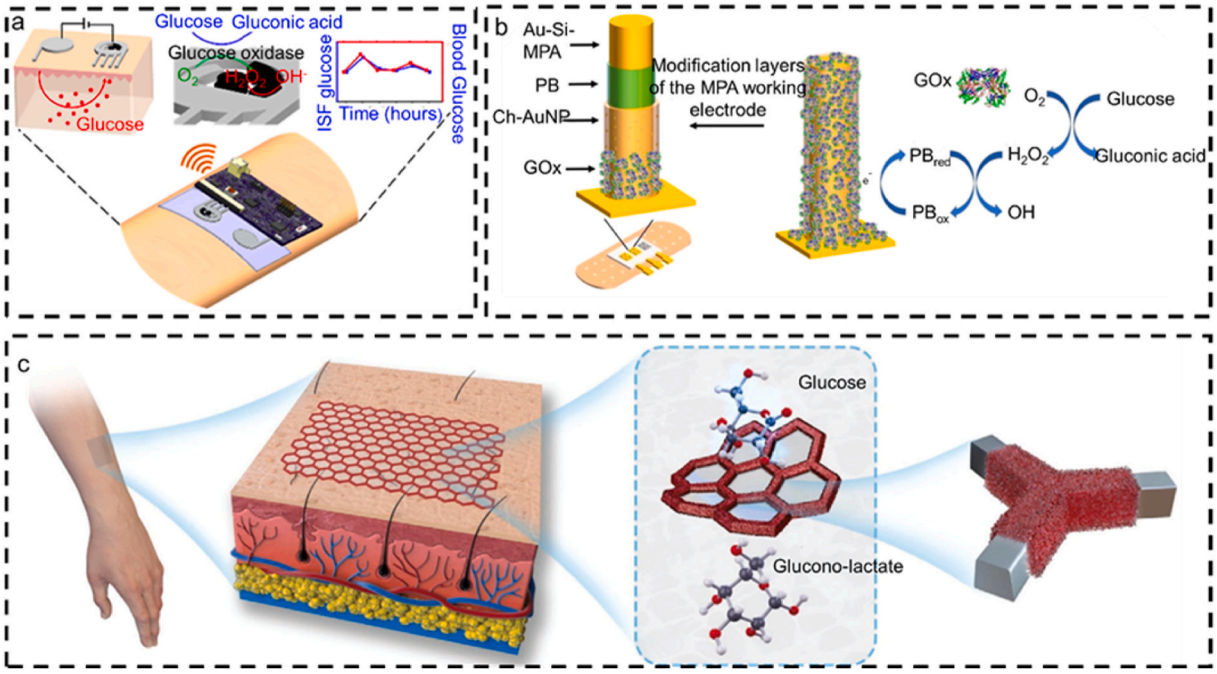


Fig. 6. Skin-interfaced bioelectronics for glucose sensing applications. (a) Schematic of the extended noninvasive glucose monitoring in the interstitial fluid using a skintronic patch. Reproduced with permission [253] Copyright 2021 American Chemical Society. (b) Schematic illustrations of the modified working electrode and the glucose detection mechanism *via* the micropillar array-based skintronic patch. Reproduced with permission [251] Copyright 2022 American Chemical Society. (c) Schematic diagram of the self-supporting Cu_2O@Ni micromesh-based skintronics for continuous glucose monitoring. Reproduced with permission [257] Copyright 2021 Wiley.

systems. For example, ML-powered VR/AR can potentially offer the user the most suitable action for a given situation *via* reinforcement learning or automatically detect and classify objects within the VR/AR world. In this regard, fully autonomous surgery using AI will likely be possible one day if we can find solutions to existing challenges, including limited access to patient data and regulatory restrictions on acquiring patient data to prevent data bleach.

The wide adoption of AI-powered skintronics has been challenged by the limited memory, battery life, and computing power of the small machinery to execute required tasks [234]. To this end, recent efforts have been made to optimize the performance, for example, by optimizing battery usage *via* ML algorithms [235] or constructing frameworks that consume low power [236]. Similarly, a recent study by Moin et al. employed real-time, in-sensor, adaptive ML to augment the clinical utility of wearable technology [237]. Typically, wearable sensors require external computing devices to perform ML model development and retrain the algorithm with new data, as they are computationally expensive. However, offline learning or training on an initial static dataset without any model updates tends to result in degraded model performance, especially when the initial training dataset does not sufficiently represent the target population. Moreover, the need for frequent retraining of ML algorithms decreases the practical utility of such a wearable sensor platform. To this end, Moin et al. used ML and screen-printed surface electromyography (sEMG) sensors to classify 13 different hand gestures while enabling in-sensor model updates *via* hyper-dimensional (HD) computing. The HD computing framework is more computationally efficient than DNN-based models as it: (1) utilizes relatively simple mathematical operations to compute cluster centroids and store calculated hypervectors in an associative memory; (2) does not require full model retraining when adding new classes; and (3) employs a nearest-neighbors approach in classifying input vectors for classification tasks. Moreover, existing vectors can be updated by simply fusing new and stored hypervectors. All hyperparameters associated with the algorithm were tuned using a publicly available dataset and subsequently trained with newly collected motion data in real-time. The results were astonishing with a classification accuracy of 97.12% with merely 4 s of in-sensor model training per gesture.

4. Skintronics for healthcare applications

Skin is the largest organ of the human body. It plays a vital role in homeostasis and thermoregulation and serves as the protective barrier against airborne toxins, pathogen diffusion, pollutants, and other harmful substances [238–242]. Although several wearable systems that monitor the biophysical characteristics of the skin have been established, they entailed bulky handheld probes or inconvenient operations and lacked real-time and continuous feedback [243]. Emerging skintronic devices can address these issues while achieving qualified assessments equivalent to clinical ones [244–248]. This section summarizes and discusses the advances in skintronics for versatile healthcare applications. In the end, we also briefly introduce the transdermal sensing platform based on interstitial fluid.

4.1. Skintronics for biosensing applications

Versatile biomarker monitoring with wearable electronics has been challenging due to several hurdles. The human epidermis is mostly curved and soft, creating imperfect contact with conventional rigid wearable devices. The need for laboratory settings (*i.e.,* bulky equipment is required to transduce the signals) also undermines the practicality of skintronics in point-of-care applications. Recent advancements in skintronics, such as device miniaturization, resolve these limitations in adopting biosensors to monitor physical and physiological parameters and metabolic biomarkers. This section highlights the recent skintronics in glucose, lactate, ion, and protein monitoring for personalized healthcare.

4.1.1. Glucose sensing

Glucose, a specific biomolecule targeted by most sensing research, collects huge attention in health monitoring with skintronics, especially

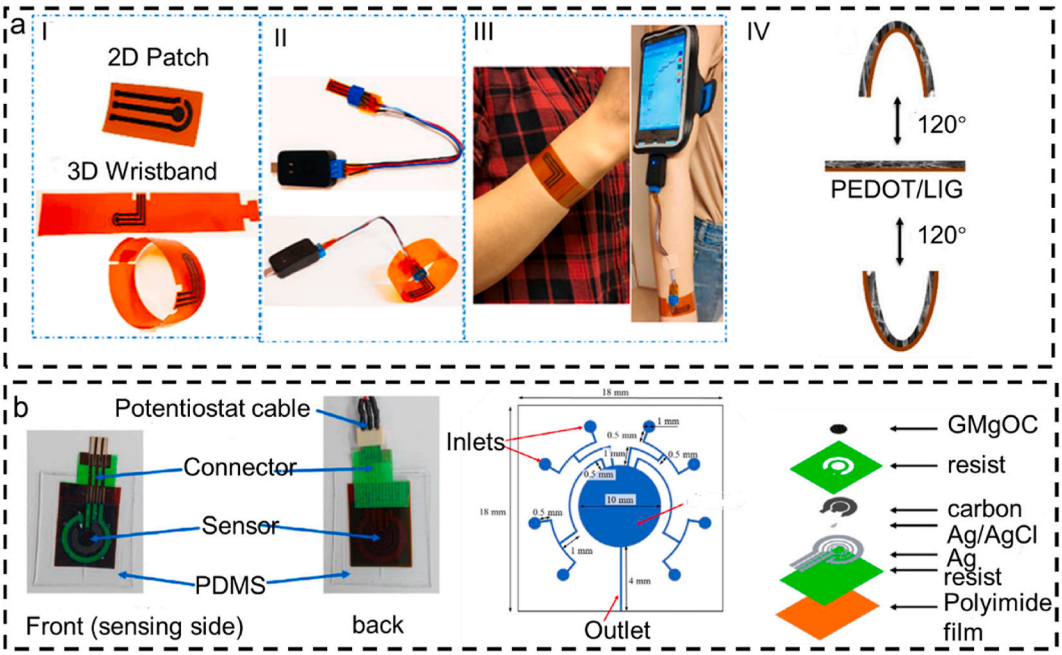


Fig. 7. Skin-interfaced bioelectronics for lactate sensing applications. (a) Digital image of the 2D patch and 3D wristband with the integrated three-electrode system for lactate sensing from the human sweat. Reproduced with permission [263] Copyright 2021 American Chemical Society. (b) Device configuration and structural revelations of the screen-printed MgO-templated carbon-lactate oxidase-based skintronic for continuous sweat lactate monitoring. Reproduced with permission [264] Copyright 2020 Elsevier.

for diabetic patients. Low-level glucose is secreted at the epidermis layer, and glucose concentration in sweat directly correlates with blood glucose level [249,250]. Detecting glucose levels in sweat thus becomes a hotspot in wearable healthcare applications. One effective strategy is to exploit the enzymatic process of glucose oxidase (GOx) and transduce the signal through electrochemical settings. In the presence of oxygen, GOx oxidizes glucose into gluconolactone, producing hydrogen peroxide (H_2O_2). The produced H_2O_2 could then get reduced at the electrode surface. This redox process generates one free electron per oxidized glucose, allowing a precise correlation between glucose level and the electron signals.

However, sweat has a lower glucose level than blood, which may affect the sensing quality. There are several ways to address this challenge. One approach is to use minimally invasive medical devices (*i.e.,* microneedles) to penetrate the epidermis layer and access the glucose in the dermis interstitial fluid (ISF). For example, Zhao et al. devised a microneedle-based biosensing platform to characterize glucose levels [251]. A three-electrode system was integrated into a silk-based microneedle patch and employed to monitor ISF glucose levels ranging from 1.7 to 10.4 mM. Such a microneedle device could sustain the sensing capability for over five weeks in artificial ISF. Another approach is to use electrochemical activity-enhancing materials. For example, Prussian blue (PB) was widely reported as a catalytic material to increase the signal intensity of the glucose biosensors [252].

In addition, carbon materials such as CNTs were integrated into the biorecognition interface to enhance detection efficiency. In PB-incorporated glucose sensors, a mixture of binding materials, such as Nafion, GOx, and glutaraldehyde, were mixed and dropped onto PB electrodes [253]. Da la Paz et al. fabricated such an electrode as a positive electrode, then incorporated a negative electrode to achieve reverse iontophoresis (IP) for continuous sweat glucose monitoring (Fig. 6a). Specifically, they applied a small current at the positive electrode, allowing the ISF glucose to be collected at the negative electrode where the glucose sensor is located. This approach concentrated the total glucose on the skin for detection, which achieved a sensitivity of 0.4 μA/mM, with a linear range of 0–22 mM of glucose.

So far, most glucose sensors discussed in this section have remained flat. Derviservic et al. reported that a three-dimensional pillar system was constructed using a micropillar array structure to improve contact efficiency with epidermal sweat (Fig. 6b) [251]. A gold-silicon-based micropillar array (Au–Si-MPA) is coated with PB and followed by the immobilization of GOx. This system reported a sensitivity of 4.7 μA/mM in a linear range of 0.05–1.4 mM glucose.

Finally, simultaneous sensing of multiple biomarkers in sweat has also been interesting. The multiplexed sensing of glucose and other physical attributes (*e.g.*, pH and temperature) were reported. As demonstrated by Wiorek et al., they integrated a fluidic device with a PB-coated carbon electrode, chitosan, GOx, and Nafion mixture [254]. In addition, two other electrode systems, one coated with H^+ selective membrane and the other modified with CNTs, were utilized to simultaneously achieve pH and temperature monitoring. This sensor achieved a 1.36 nA/μM sensitivity in 0–200 μM of glucose. Meanwhile, it has excellent pH and temperature sensing performance with a sensitivity of −52.9 mV/pH and −4.4 kΩ/°C.

Recent research reported electrochemical non-enzymatic detection of glucose [255]. Han et al. polymerized L-lactic acid on the platinum-interdigitated electrode [256]. Such a device showed a LOD of 0.19 nM and a sensitivity of 0.387 μA/μM in the dynamic range of 0.001–1000 μM glucose. Another system reported by Li et al. developed a nickel mesh coated with Cu_2O to oxidize the glucose (Fig. 6c) [257]. This sensor showed a 15,240 μA/cm sensitivity in 50 nM-1.07 mM glucose. Both non-enzymatic sensors showed comparable performance with enzymatic biosensors.

Other sensing methodologies, such as optical approaches, were also reported to detect sweat glucose. For example, Vaquer et al. developed a colorimetric-based flow sensor [258]. A mixture of GOx, horseradish peroxidase (HRP), and 3,3′,5,5′-tetramethylbenzidine (TMB) was coated onto a paper strip. When sweat enters the inlet of the flow strip, the glucose reacts with the GOx, producing H_2O_2, which facilitates HRP to oxidize the TMB and then generates a bluish color. This wearable system reported a LOD of 0.01 mM with a dynamic range of 0.01–0.15 mM glucose. Except for HRP, MnO_2 with graphene oxide was used to facilitate the oxidation of TMB *via* H_2O_2 produced by GOx [259]. This colorimetry platform showed a dynamic range of 0–500 mg/dL. This

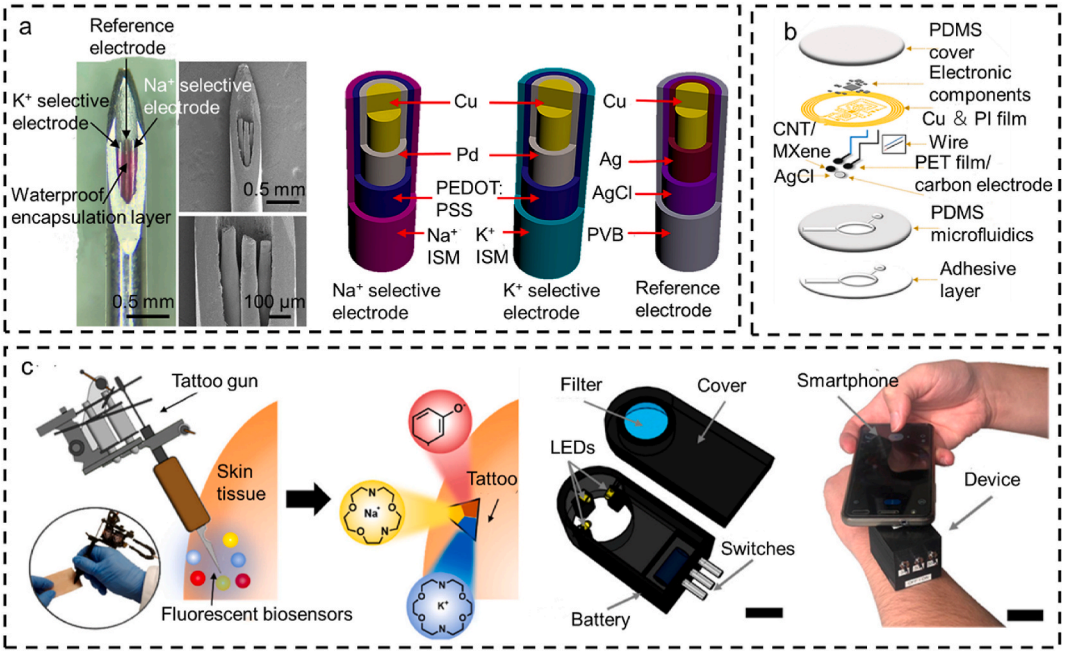


Fig. 8. Skin-interfaced bioelectronics for ion sensing applications. (a) Microneedle-integrated skintronics based on a potentiometric detection system for simultaneous Na^+ and K^+ monitoring in ISFs and the schematic illustration of multiple functional layers of the sodium ion-selective electrode, the potassium ion-selective electrode, and the reference electrode. Reproduced with permission [326] Copyright 2021 American Chemical Society. (b) Schematic illustration of the exploded view of the microfluidic-based skintronic device for the on-site ions monitoring (*e.g.,* K^+, Ca^{2+}, Zn^{2+}, and Na^+) in human perspiration. Reproduced with permission [327] Copyright 2021 Elsevier. (c) Fluorescent electronic tattoo platform for monitoring non-invasive electrolytes (H^+, Na^+, and K^+). Reproduced with permission [276] Copyright 2020 Elsevier.

work also established an intelligent delivery system based on the microneedle. Once the glucose threshold reaches a predesignated level, insulin will be released automatically.

Overall, most of the reported skintronics with the capability of sweat glucose sensing still employ GOx to facilitate the production of a detectable electrical signal. Although the enzyme-based monitoring method is cost-effective and easily operable, it is highly susceptible to protease degradation and requires a temperature-controlled storage environment. Therefore, further investigation into non-enzymatic glucose detection is desirable.

4.1.2. *Lactate sensing*

Aerobic metabolite sensing, such as lactate, holds promise for clinical applications [260]. Analogous to glucose sensing, lactate detection in sweat can also be achieved through enzymatic reactions using lactate oxidase (LOx). In the presence of oxygen, LOx oxidizes lactate and produces pyruvate and H_2O_2. H_2O_2 gets reduced at the electrode interface, generating a detectable electrical signal.

In one example, LOx was immobilized *via* electrostatic adhesion between the negatively charged protein and the positively charged ZnO coated on a polyamide substrate [261]. This device was reported to have a LOD of 1 mM with a dynamic range of 1–100 mM lactate. In another study, simultaneous detection of glucose and lactate was reported. In contrast to the electrostatic adhesion, β-cysteamine was used as a surface linker on the gold surface with microstructures grown by immersion deposition [262]. Then GOx for glucose sensing and LOx for lactate sensing were mixed with bovine serum albumin, and poly (ethylene glycol) diglycidyl ether was added to the surface to increase enzymatic stability. This sensor showed a LOD of 7 μM with a detection range of 25–250 μM glucose, while it displayed a LOD of 54 μM with a detection range of 1–25 mM lactate. One continuous trend seen throughout biosensor fabrication is the addition of carbon nanomaterials at the recognition interface to enhance the electrical signal. In contrast to previous studies using pre-formed nanomaterials, Meng et al. presented a laser-irradiation approach to pattern graphene on polyimide substrates [263]. The resulting laser-irradiated graphene (LIG) is reinforced using poly (3,4-ethylene dioxythiophene) to create a flexible and conductive base. A PB and LOx mixture was then deposited onto the PEDOT/LIG substrate (Fig. 7a). Such a device could detect H_2O_2 and lactate with LOD of 2.5 μM and 6.8 μM. For H_2O_2, the sensitivity was 24.33 μA/mM in the range of 0.01–1.76 mM. The sensitivity for lactate was 11.83 μA/mM in the range of 0.01–1.35 mM.

Wearable sensor systems combined with microfluidics can also be seen in skintronics for monitoring sweat lactate. This strategy can resolve uneven contact between the sensor and the skin. For example, a microfluidic system integrated electrochemical lactate sensor was fabricated to have inlets to collect sweat and a central chamber housing an electrochemical lactate sensor (Fig. 7b) [264]. The electrodes were printed on a polyimide film, and then the working electrode was modified using MgO materials to serve as a signal-enhancing element. Their sensor achieved a LOD of 0.3 mM, with a sensitivity of 36.2 $\mu A/cm^2/mM$. Another advantage of using a sweat-collecting fluidic device is that it allows accurate sweat sampling directly to the sensing area, which increases the sensing efficiency of biorecognition elements on the sensor.

Apart from the utilization of microfluidic units, the inherent capillary phenomenon of paper was exploited to monitor sweat lactate. Ardalan et al. reported a multiplexing sensor on paper substrates to detect glucose, lactate, and pH [265]. A variety of fluorescent molecules were coated on the paper sensor. The concentration of the target molecules could be quantified by measuring fluorescence intensity. The sensor showed a LOD of 7 μM in the linear range of 10–250 μM for glucose and a LOD of 0.4 mM in the linear range of 1.0–12.5 mM for lactate. More practical approaches for detecting small molecules in sweat are still emerging [266–268]. Such exciting improvements seen throughout wearable sweat lactate monitoring are encouraging and inspiring. Skintronic devices are expected to become a significant player in lactate monitoring for supporting clinical pre-screening processes.

4.1.3. Ion sensing

The second biomarker-of-interest for monitoring using skintronics is ions in sweat. Common ions, including H^+, Na^+, K^+, and Cl^-, can be abundant [269]. Their concentration can reflect athletic ability and human health [270]. Hence, monitoring ion concentration in sweat can provide a promising diagnostic tool. For example, Li et al. developed an ion-selective micro-electrode integrated needle system to detect Na^+ and K^+ [271]. They coated a copper-platinum core-shell electrode with PEDOT: PSS and an ion-selective membrane. These electrodes are coated onto a needle-like structure. The electrode-integrated microneedle penetrates the skin to access the subepidermal ISF (Fig. 8a). The sensor showed a sensitivity of 56.08 mV/mM, in the linear range of 0–200 mM of Na^+. For K^+, a sensitivity of 50.03 mV/mM and a detection range of 0–15 mM were reported. However, this system still requires penetration of the epidermis layer. Though minimally invasive, it may result in discomfort. In addition, an external power source and additional wire connection are required to operate the sensor.

Another study resolved these issues by integrating a near-field communication (NFC) component with a radio frequency (RF) antenna to achieve power and signal transduction [272]. In addition, a PDMS microfluidic sweat sampling system can be integrated to achieve maximal sample collection efficiency for ion detection. As for the ion sensor, an electrode was coated with CNT and MXene ($Ti_3C_2T_x$). CNT can enhance the layer-to-layer stability of the MXene mixture, and MXene can increase electrocatalytic activity on the electrode (Fig. 8b). Finally, a layer of valinomycin was coated on the electrode to achieve a high K^+ selectivity of the electrode at 63 mV/dec in the linear range of 1–32 mM. This wearable sensor does not require an external power source [273]. Another wearable sensor detailed an interesting fiber-typed Na^+, Cl^-, and pH multiplexing sensor [274]. In this work, gold nanowire (AuNWs) was impregnated with styrene-ethylene/butylene-styrene (SEBS), plated with gold, and coated with an ion-selective membrane. This sensor reported a sensitivity of 59.2–57.4 mV/dec in the linear range of 0.1 mM-0.1 M towards Na^+. Their fiber sensor demonstrated only a 2.1% increase in resistance under 200% strain.

Besides potentiometric sensors, field-effect transistor (FET) is also employed in ion sensing. In recent work, CNT-based FET has gained considerable attention due to its stable electrocatalytic activity and tolerance in a high-salt environment [275]. An ionophore calixarene was doped into a polyvinylchloride (PVC) membrane covering the FET [24]. The sensor correlated the decrease in threshold voltage to Na^+ concentration. They reported a sensitivity of 71.7 mv/dec in the detection range of 0.1–100 mM towards Na^+. So far, most ion detection has been achieved on devices fabricated on polymer or gels. In one study, the author presented a unique ion-sensing approach. They incorporated ion-sensitive fluorescence dye into the epidermis *via* tattoo and transduced signal using a handheld fluorescence detector (Fig. 8c) [276]. The presence of ions changes the fluorescence of these dyes, enabling ion detection through a compact handheld fluorescence detector. Their tattoo ion sensor demonstrated a detection range of 6.6–7.6, 100–175 mM/L, and 2.0–6 mM/L towards pH, Na^+, and K^+, respectively.

In summary, ion sensing is still a hot topic in skintronics. Some interesting approaches to sensor fabrication are emerging or remain to be further investigated. Various materials, platforms, and systems are adapted into ion sensors at the skin interface. This is an exciting trend, especially in comparison with the biosensors for metabolite monitoring, which demonstrate a huge disproportion utilizing enzymatic reaction at the detection interface, as described in the previous sections.

4.1.4. Protein sensing

In contrast to the monitoring of metabolites and ions, skintronics that target epidermal proteins are less abundant. This is foreseeable since a higher population of proteins is found in the proximal fluids. However, several significant biological proteins that hold clinical importance can be found in sweat related to cancer progression, inflammation, and infection. An electrochemical-based closed-loop system was reported to monitor inflammation biomarker interleukin-1β and C-reactive in the sweat [277]. Also, a FET-based sensor was reported to detect bacterial infection-related proteins in the sweat. For example, a FET was fabricated using a graphene-Nafion composite film to detect cancer biomarker cytokines interferon-gamma (INF-γ). Compared to the previous report, this study utilized INF-γ specific aptamers immobilized on graphene-Nafion. When INF-γ binds to the immobilized aptamer, the gate voltage changes to achieve detection, their system reported a detection limit of 740 fM in the range of 0.015–250 nM. Indeed, the system's performance is remarkable. One critical issue that needs to be considered is sweat has high ion concentration. Hence the performance of INF-γ-FET during thermal regulation still needs to be evaluated. In another cytokine detection skin sensor, a microneedle structure was utilized. As demonstrated by Song et al., they fabricated a microneedle on an Al_2O_3 substrate to detect human interleukin 8 (hIL8) [278]. The needle is layered using gold to establish electrical conductivity, and they also imprinted microwell at the needle tip for anti-cytokine hIL8 antibody functionalization. This needle structure is inserted sub-epidermally to access the ISF. In the presence of hIL8, it binds with the immobilized antibodies, changing the impedance. This sensor showed a detection range from 62 pg/mL to 539 ng/mL.

Skintronics for biosensing is an ongoing research topic. As flexible conductive materials continue to advance, wiring issues that previously hindered biosensor fabrication on soft materials will become possible. Also, as seen through the work of Zhang et al. [272], incorporating NFC and antenna structure, signal transfer, and powering elements required to run skin biosensors can become wearable and/or portable. This further increases the point-of-care potential.

4.1.5. Strain-insensitive electrochemical biosensing

Biological tissue, such as skin, is soft and shape-adaptive with complex topologies. To this end, researchers focus on adapting intrinsically stretchable and conductive materials to perform designated electrochemical reactions for biomarker biosensing. Specifically, it includes conductive polymer (for example, PEDOT: PSS), metal nanowires, and CNTs, simultaneously serving as electron transport (conduction) and charge transfer (electrochemical reaction) materials of electrochemical biosensors. However, when these mechanically matched biosensors are interfaced with the tissue, a fundamental challenge is that tissue-induced deformation leads to conductivity degradation of electrode materials [279,280]. As a result, obtained electrochemical biosensing signals are distorted, and thus, biomarker concentrations are misinterpreted in envisioned clinical settings and biological studies.

The electrochemical signal distortion originates from the conductivity of the electrode variation under strain dominating over solution resistance and electrochemical impedance at the electrode-biofluid interface. To be more specific, the electrochemical signal pathway can be modeled as three significant components in series based on the Randles model: 1) a resistor to represent biofluid/tissue; 2) a resistor in parallel with a capacitor illustrating faradaic and non-faradic process as two independent charge transfer pathways at the electrode-biofluid interface and 3) a resistor to depict electron transport in a solid-state conductor. The solution resistance is generally on the order of 100–1000 Ω for a typical mm-scale bioelectrode due to the high intrinsic conductivity of ion-abundant biological fluids (10^{-2} to 10^{-1} S/m) [281,282]. Electrochemical impedance could be as low as ~10 Ω/cm^2 but also increase several orders of magnitude depending on interfacial materials, applied voltage amplitude, and frequency of desired electrochemical reactions and analytes concentration. Consequently, to minimize signal distortion by strain, resistivity, and strain-sensitivity of conductive materials should be as low as possible to ensure that strain-induced resistance increase could be negligible compared with the solution resistance and electrochemical impedance [283]. To this end, AgNWs and liquid metal are the ideal electron transport materials

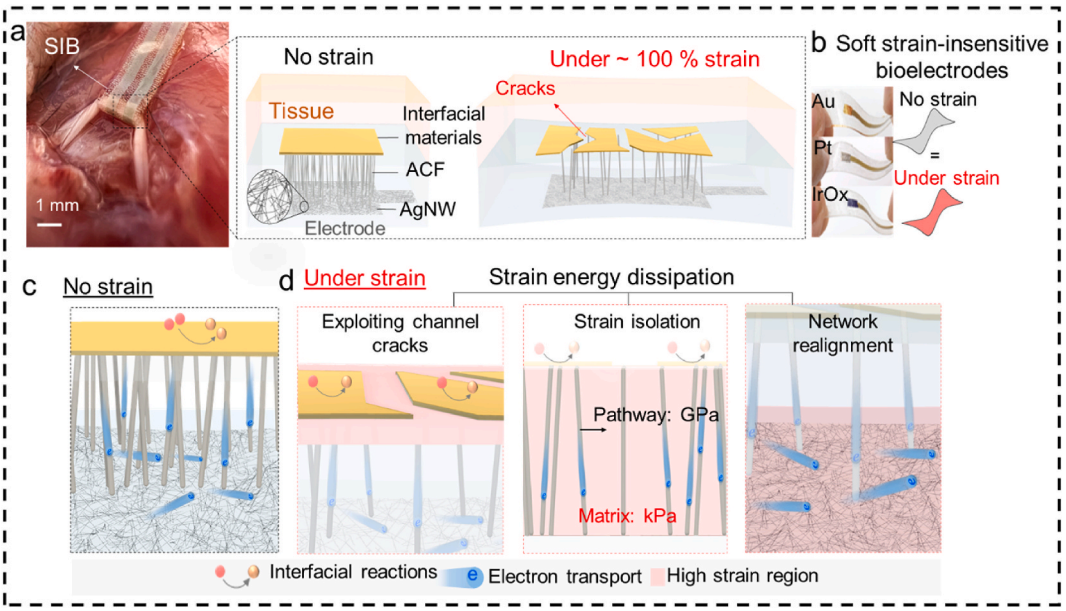


Fig. 9. Strain-insensitive bioelectronics (SIBs) for electrochemical biomarker sensing. (a) Illustration of the SIB under strain when interfacing with biological tissue. (b) SIBs featuring different brittle noble metals for high-fidelity sensing under strain. (c) Schematic of charge transfer and out-of-plane/in-plane electron transport processes within SIB. (d) Illustration of SIB's strain energy dissipation mechanism. Reproduced with permission [284] Copyright 2022 Springer Nature Group.

to construct strain-insensitive sensing bioelectrodes due to their tissue-like stretchability and metal-like conductivity. Accordingly, as shown in Fig. 9a–b, Zhao et al. report a strain-insensitive bioelectrode (SIB) consisting of 1) AgNW as the bottom in-plane electron transport layer, 2) an anisotropic conductive adhesive as the intermediate out-of-plane electron transport layer and 3) channeling-cracked brittle, thin film as the charge transfer layer [284]. In the devised configuration (Fig. 9c-**d**), each cracked fragment of the brittle film remains electrical connection with the electron transport layers underneath. Therefore, the overall electrochemical active surface area is unchanged. As the out-of-plan conduction layer, anisotropic conductive adhesive renders strain-insensitive vertical electrical transport, since the rigid conductive microparticles (~GPa) are strain-isolated by the surrounding soft matrix (~KPa) The AgNWs in the bottom layer form in-plane conductive percolation networks. The network is strongly bonded with the soft polymer substrate *via* in-situ crosslinking. Accordingly, it enables AgNWs to remain high conductivity under deformation *via* network realignment to dissipate strain energy. To enable a broad range of biomarkers sensing capabilities, the interfacial layer of SIB can be replaced by different thin films (Au, Pt, Ir, and C) based on the envisioned electrochemical reactions. Coupled with a diverse set of electrochemical techniques such as voltammetry, amperometry, and electrochemical impedance spectroscopy, SIB presents strain-insensitive and high-fidelity sensing capabilities of multiple biomarkers of interest.

4.2. Skintronics for wound healing

Wound healing is a complicated but normal biological process after an injury, which involves several physiological events, including hemostasis, inflammation, proliferation, and remodeling. [285,286] The healing process can be affected by various factors and may lead to non-healing wounds, significantly burdening patients and healthcare systems [287]. For example, diabetes was reported to impair the late inflammatory response during wound healing, and it can cause the wound to remain non-healed for months or even years [288]. Therefore, monitoring wound healing/trauma repair is essential [289–296]. Among various bioengineering tactics, skintronic platforms are increasingly attractive for wound management [34,297–300]. For example, Ochoa et al. developed an on-wound patch for oxygen sensing and delivery to overcome the suboptimal oxygenation of the wound bed [299]. The skintronic device could lift oxygen concentration in a gel substrate by 13% (5 ppm) within 1 h and provide luminescence-based oxygen sensing ability with a range of 5–26 ppm. Besides, the flexible oxygenation microsystem was easily fabricated in a layer-by-layer method. The *in vivo* studies demonstrated its biocompatibility, biostability, and ability to increase oxygen levels to the clinically-graded requirements [299].

Another interesting example of the skintronics used in wound management is named "drawn-on-skin" (DoS) electronics (Fig. 10a) [34]. DoS electronics have provided a unique approach for point-of-care treatments with advantages over conventional bioelectronic devices, such as facile fabrication processes, multifunctional sensing, and motion artifacts-free. The DoS electronics were generated *via* a ballpoint pen to pattern the skin surface with conductive inks. The DoS electronic devices used Ag-PEDOT: PSS composite, poly (3-hexylthiophene-2,5-diyl) nanofibrils (P3HT-NF), and ionic gel as the conductive, semiconducting, and dielectric inks, respectively. With its customizability to wound shapes and sizes, DoS was able to facilitate wound healing/recovery *via* pulsed electrical stimulation. Therefore, establishing interfaced DoS electronics with a built-in miniature electrical stimulator on injured skin is promising regarding on-the-go and personalized treatments [34].

4.3. Skintronics platform for sleep monitoring

Sleep is vital to restore the whole body from fatigue and affects human health and quality of life [301,302]. Poor sleep quality may lead to severe sleep disorders causing increased stress, precarious emotions, weak memory, and poor brain functions [303–306]. Polysomnography (PSG) is a conventional clinical method for sleep analysis, but the need for complicated procedures, professional assistance, and undesired sleep interference makes it impractical for at-home sleep monitoring. In comparison, galvanic skin response (GSR) is a good indicator for sleep monitoring. Kim et al. reported an integrated skintronic device for GSR measurements featuring conformal lamination without using external tapes/straps (Fig. 10b) [307]. The mechanical properties and signal processing in sleep and wake states have been evaluated. Attaching the

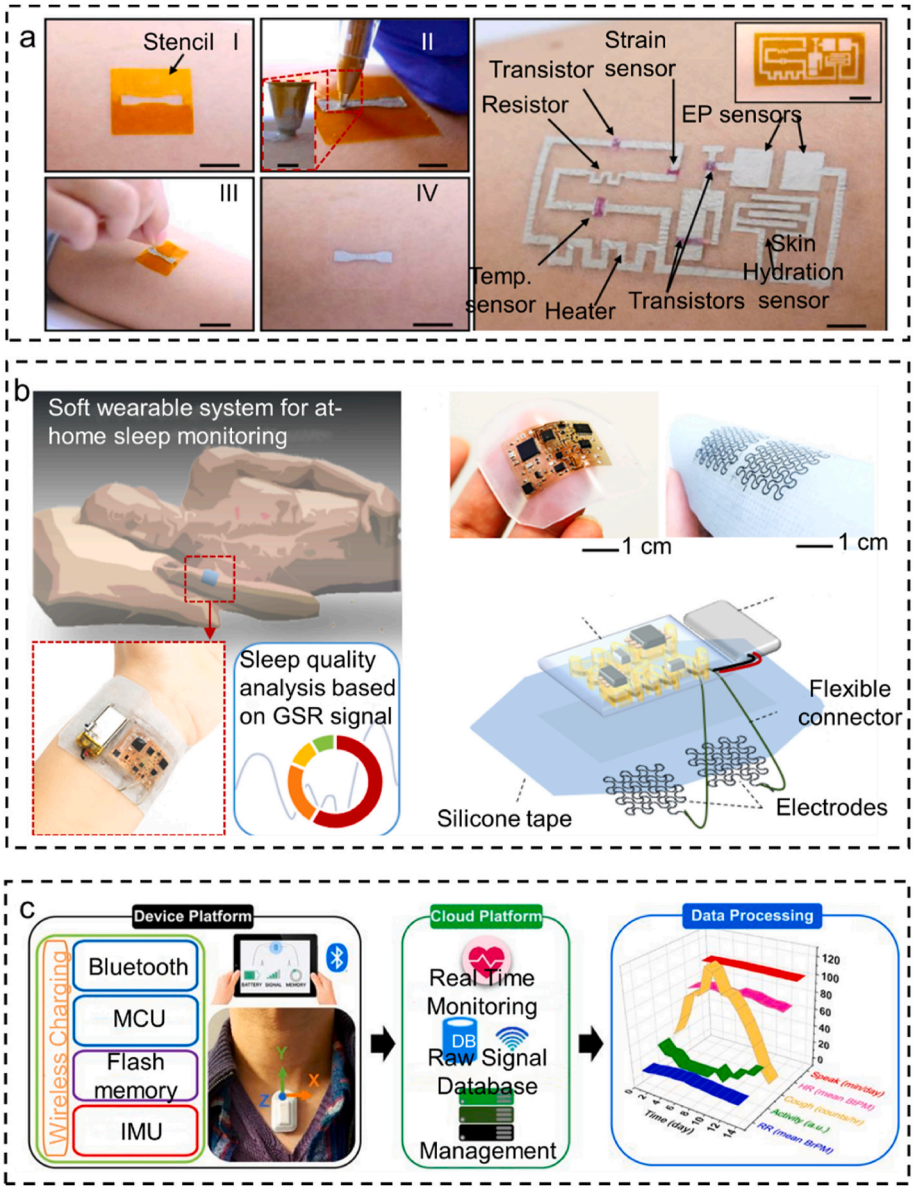


Fig. 10. Skin-interfaced bioelectronics for personalized healthcare applications. (a) The ultra-conformal drawn-on-skin (DoS) electronics with the conductive Ag-PEDOT: PSS ink for wound healing applications. The final DoS skintronics includes a resistor, transistors, heater, and electrophysiological, temperature, strain, and skin hydration sensors. Reproduced with permission [34] Copyright 2020 Nature Group. (b) A soft skintronics device to record galvanic skin response for sleep monitoring. This integrated device comprises an elastomer, a pair of graphene electrodes, a circuit, and a battery on reusable silicone tape. Reproduced with permission [307] Copyright 2021 MDPI. (c) A skintronic platform for the automated, multiparametric monitoring of respiratory biomarkers and vital signs for tracking SARS-COV-2. The overall operational flow of the system consists of a device, cloud, and data processing platforms. Reproduced with permission [314] Copyright 2021 PNAS.

device to the wrist surface could generate high-quality GSR signal recording to monitor the sleeping status of the users [307].

4.4. *Skintronics-enabled VR/AR for healthcare applications*

The future of VR/AR aims to offer a fully immersive setting where individuals can experience sounds, images, and sensations [308]. The mechanoreceptors, distributed throughout the human dermis, form the basis of our physical interactions with the world. However, the integration of mechanoreceptors-integrated VR/AR platforms has not been profoundly studied. Rogers's group engineered a skintronic platform to communicate information *via* spatiotemporally programmable patterns of localized mechanical vibrations (Fig. 11a) [309]. The materials and device design, with optimized system layouts for controllable wirelessly-powered haptic interfaces, have been systematically studied, extending VR/AR-enabled audiovisual experiences through a remote computer system. One scenario was the virtual interactions between a girl and her grandmother *via* social media. When the girl touched the screen, her grandmother, skin-mounted with two VR-integrated skintronic devices, could experience a continuous wave of vibratory excitation from her forearm down to her hand, corresponding to the movements of her granddaughter's fingertips (Fig. 11b, top). Another scenario was applied to these robotic prosthetic devices designed for amputated individuals with a skintronic-empowered VR device on the residual limb (Fig. 11b, middle). When grasping objects, the wearer could recognize the object's shape and have a virtual haptic sensation on their upper arm. Besides, the device enhanced the haptic engagement of a gaming wearer by placing the device at different body locations to receive corresponding haptic feedback during a combat game (Fig. 11b, bottom) [309].

Many studies have focused on developing artificially-reconstructive human senses, such as visual, auditory, and vibrational, *via* VR/AR-enabled applications. However, most rigid and bulky products lead to unsatisfying user conformability [310]. Therefore, developing VR/AR-enabled skintronics platform for haptic modes by intimate and conformal contact with the skin is appealing to overcome these issues, including inflexibility and inefficient temperature conduction with the human epidermis. For example, most electronics were designed for heating-dependent processes, while the cooling mode has not been extensively explored. Because the single temperature mode does not reflect a human's actual thermal sensation process, the cooling and heating functionalities are expected to realize the thermal-haptic

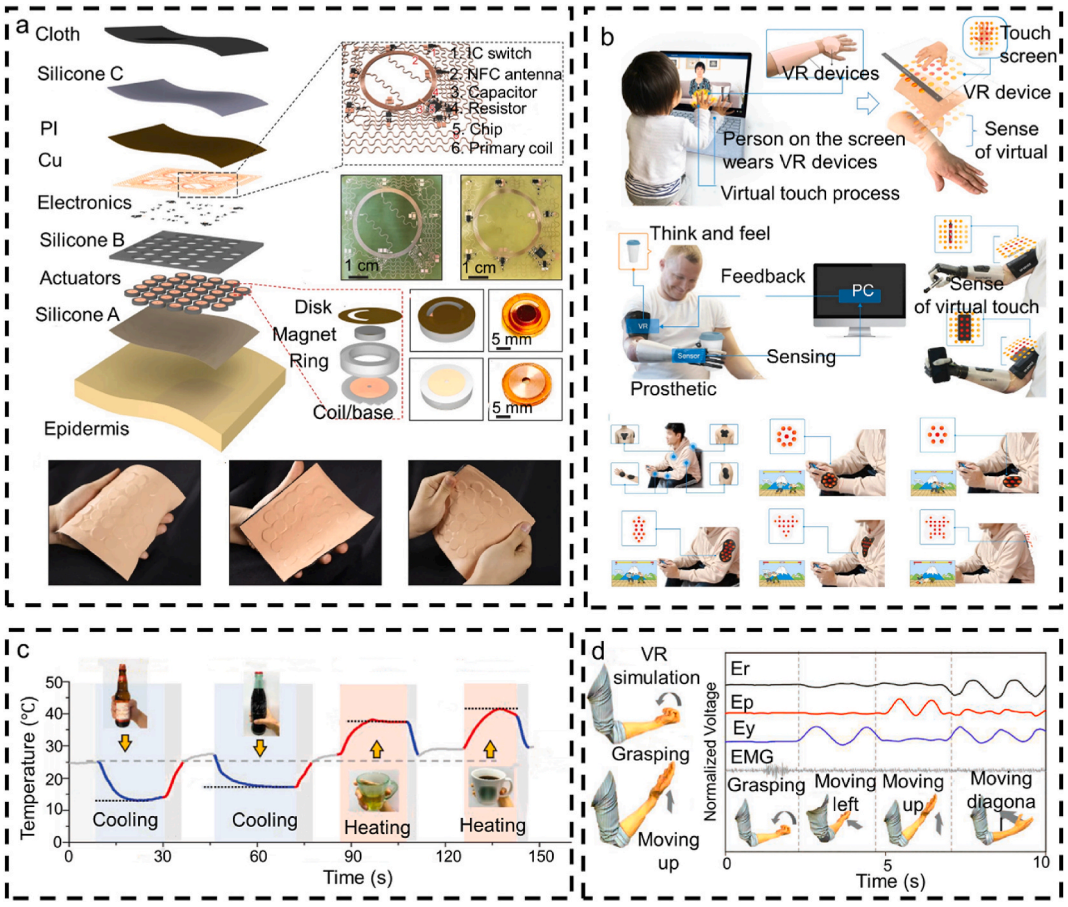


Fig. 11. VR/AR-enabled Skin-interfaced bioelectronics-enabled for healthcare applications. (a) Design and architecture of a VR-enabled skintronic system that integrates the NFC and flexible circuit. Reproduced with permission [309] Copyright 2019 Nature Group. (b) Examples of applications of the VR-enabled skintronics system: social media applications (top), prosthetics applications (middle), and gaming applications (bottom). Reproduced with permission [309] Copyright 2019 Nature Group. (c) The graph shows temperature change based on the virtual scenario of touching various objects to confirm the reconstruction of artificial thermal sensation in virtual reality. Reproduced with permission [328] Copyright 2020 Wiley. (d) The developed skintronics used to control the virtual hand in real time, and the right part summarized the measured gyroscopic and EMG signals. Reproduced with permission [201] Copyright 2021 Wiley.

sensation changes [311,312]. In this regard, Lee et al. constructed a skintronic platform based on a thermo-haptic (STH) device, enabling the artificial heating and cooling thermal sensation of the skin [311]. In addition, the rapid temperature changes between heating and cooling modes without any unintended parasitic thermal residues facilitated the accuracy and controllability (Fig. 11c) [311]. Furthermore, to prove the feasibility of this STH device in personal thermal management, the authors carried out thermal change experiments *via* localized cooling and heating. The result showed that the temperature change between the up-and-down was around 13 K, indicating its effectiveness in provoking warm/cold feelings. Moreover, using a reverse mode of the STH enabled a rapid temperature reversal to room temperature after touching cold and warm objects without thermal residues. This sensitive response provided accurate thermal sensation, generally absent in other single-mode heaters/coolers. Besides, the independent operation of devices affixed on three different fingertips illustrated that the individual STH device could be used to provoke a different localized thermal sensation on each finger. Altogether, the device characteristics can be integrated with VR to imitate real-time thermal sensations in virtually-designed settings and can provide a potential medical approach to train and restore patients' sensational ability [311].

Another example was reported by Yao et al., who investigated the predictive and electro-mechanical responses of a skintronic platform with a wide pressure range of more than 100 kPa [313]. The association of geometry designs, material properties, and single-pyramid performance was comprehensively studied using simulations and real-life empirical experiences. With further optimization of the sensitivities to match the tissue stiffnesses, the AR-interfaced platform was demonstrated for training [313]. Similarly, an AR/VR skintronics platform adopted the innovative "*in situ* and in operando adaption (SOA)" method to address unnecessary waste and disposal issues and achieved various user-oriented specifications (Fig. 11d) [201]. With the addition of VR/AR functions, remote control of virtual objects with skintronics became available.

5. Conclusions, commercial perspectives, and clinical promise

Emerging skintronic devices are promising platforms that offer a range of capabilities, from the analysis of clinically-relevant vital signs and daily health/fitness management to the future AR/VR-enabled Internet of Things (IoT). Compared to conventional and current commercially available wearables (*i.e.,* Apple Watch and Fitbit wristband), skintronics are thin, soft, and comfortable. Intensive research in skintronics has been accomplishing continuous and real-time health monitoring in a non/minimally invasive manner. The skintronics, combined with advanced ML technologies, have also shown applicability in the health monitoring of COVID-19 patients and have been envisioned to provide backup alternatives to current medical devices [314]. Skintronics have also demonstrated immense potential for military use. Furthermore, a wireless integrated skintronic platform holds enormous promise to provide practical medical guidance and timely intervention for monitoring neonatal physiological status [315].

Table 1
Summary of the skintronic devices for various biomedical applications.

Sensor Type	Design Principles	Features	Application	Detection Range	Ref.
T&P Sensor	Full-body T&P monitoring for personalized healthcare	Being thin and soft with the ability to measure physiological signals precisely and continuously; NFC-enabled power delivery and data transmission	Full-body pressure and temperature mapping	Temperature (23–35.5 °C); pressure (15–60 mmHg)	[329]
T Sensor	Skin thermal properties-related clinical interest	Being soft, flexible, and battery-free; temperature and skin hydration monitoring; NFC-/RFID-enabled wireless powering and communication; low-cost and mass production for clinical translations.	For multimodal thermal characterization of human skin	N/A	[247]
E-Tattoo Sensor	Design based on the Gr/SF/Ca^{2+} ternary system	Being soft, flexible, stretchable, mechanically robust, and self-healable	Strain, humidity, temperature, ECG, and temperature sensing	Strain (0–70%) Humidity (11.3–85.1%); Temperature (20–50 °C)	[322]
Microfluidics-enabled Sensor	Sweat-based microfluidics-enabled epidermal electronics design	Waterproof, epidermal, microfluidic, and electronic systems with excellent skin conformability to enable capture, storage, and analysis of sweat; quantitative *in situ* measurements of local sweat chloride concentration, local sweat loss, and skin temperature	Sweat collection and biomarker detection for aquatic and dryland athletes	N/A	[330]
T&P Sensor	Evaluation for precise form-fitting of prosthetic sockets	Soft, 3D designed millimeter scaled pressure sensor integrated into a flexible, battery-free, wireless platform with a built-in temperature sensor; noninvasive, wireless monitoring, demonstration of clinical applications in prosthesis users	Multimodal sensing at the skin-prosthesis interface	N/A	[331]
Glucose Sensor	Iontophoretic-biosensing platform for glucose level monitoring	An easy-to-wear flexible tattoo device of reverse iontophoretic extraction of interstitial glucose; detections of micromolar glucose in the presence of common interfering chemical species	On-body noninvasive glucose monitoring	Glucose (0–100 μM)	[198]
Lactate Sensor	Immobilization of lactate oxidase	Real-time noninvasive lactate sensing in human perspiration	Real-time continuous monitoring of sweat lactate dynamics during exercise	Glucose (1–20 mM)	[332]
Alcohol Sensor	Induced sweat alcohol sensing	Integration of an iontophoretic-biosensing tattoo with flexible wireless electronics; being a highly selective and sensitive response to ethanol; Wireless data transmission in real time	Noninvasive monitoring of alcohol consumption in practical settings	Glucose (0–36 mM)	[333]
Electrolyte Sensor	Introduction of the ion-selective potentiometric cell in a temporary-transfer tattoo platform	Unaffected analytical performance with stringent mechanical stretching or bending	Ammonium monitoring for detecting metabolic disorders	10^{-4} M–0.1 M with detection limits of $10^{-4.9}$ M	[334]
Electrolyte (Na^+) Sensor	Skin-worn non-invasive tattoo sensor	Real-time monitoring of Na^+ in human perspiration with a rapid near-Nernstian response with negligible carryover effects; good resiliency against various mechanical deformations; wireless transmission from the skintronics to mobile devices	Sodium detection	0.1 mM–100 mM	[335]
BMI Sensor	Solution to designing a portable and ergonomic EEG system	Flexible with a set of dry electrodes and circuit for wireless, real-time EEG monitoring; improved performance in detection of evoked potentials due to significant reduction of noise; electromagnetic interference; high information transfer rate	Designed for a broad range of assistive devices, smart-home systems, and neuro-gaming interfaces	N/A	[336]
E-Tattoo	Sub-micrometer-thick transparent GET	Filamentary serpentine mesh fabrication *via* a cost- and time-effective "wet transfer, dry patterning" method; being transparent, flexible, stretchable, and conformable; low impedance	EEG, ECG, EMG, skin hydration, temperature monitoring	N/A	[60]
SCG Sensor	Alternative to traditional rigid and non-stretchable SCG wearables	Ultrathin and stretchable SCG tattoo sensors with filamentary serpentine mesh; effectively reduced motion artifact	SCG measurements	N/A	[337]
Multifunctional Sensor[a]	All-purpose E-tattoo with the personalized adaptability	Modular and reconfigurable E-tattoos; adaptable functional units (NFC); being rapidly prototyped *via* a digital and cost-effective fabrication process	ECG, SpO_2, HR, temperature, and skin hydration monitoring.	N/A	[200]
Multifunctional Sensor[a]	Introduction of 3D architectures without 2D engineering	Self-assembled 3D network designs; open-mesh, interconnect networks of helical micro-coils by deterministic compressive buckling; in compact geometries, with active components and sophisticated functionality	ECG, EMG, EOG monitoring	N/A	[338]
Microfluidics	On-skin integration of the soft, textured, curvilinear, and time-dynamic surfaces	Ultralow modulus, highly stretchable systems with assembled high-modulus, thin, conformable; being laminated onto the skin surface for physiological monitoring in a wireless manner	On-skin temperature and ECG detection	N/A	[339]
Multifunctional Sensor[a]	Solution to the interfacing of soft and hard electronics	Soft and rigid devices are fabricated on separate substrates; real-time, continuous monitoring with high sensitivity	On-skin ECG, HR, and temperature detection	Covering the human body temperature (32–37 °C)	[340]
		Highly flexible sensors with high sensitivity; actuators			[88]

(*continued on next page*)

Table 1 (*continued*)

Sensor Type	Design Principles	Features	Applications	Detection Range	Ref.
Biopotential Sensor	On-skin patterning of conductive LM		EEG, ECG, and EMG monitoring	Tensile length ranging from 90 to 105 mm	
Multifunctional Sensor[a]	2D dichalcogenide-based E-tattoos	$PtTe_2$ tattoos with four times lower impedance and 100 times lower R_s compared to graphene tattoos	Advanced HMI; ECG, EMG, EEG, EOG, and temperature monitoring	N/A	[52]

Sensor Type	Design Principles	Features	Applications	Detection Range	Ref.
Ventricular shunt detection sensor	Direction-dependent heat transport	Thin, flexible, epidermally adherent, and wireless communication	Detecting shunt malfunctions in patients	N/A	[341]
Oxygen Sensor	Measurements of the phosphorescence decay rate	Cost-effective, mass production, and highly flexible oxygen monitoring in the wound bed	Oxygen generating and delivery for accelerating SWH	O_2 (5–26 ppm)	[299]
Multifunctional Sensor[a]	Conductive ink	Multifunctional designs for strain, temperature, hydration, and heat, ECG/EP monitoring with immunity to motion artifacts	Accelerating SWH	Stretchability (30%); temperature (20–45 °C)	[34]
Multifunctional Sensor[a]	Pencil-drawn graphite pattern	Cost-effective, disposable, and reconfigurable; multifunctional designs for various physical signal and biomarker detection	Temperature, ECG, EMG, heart/respiratory rates, sweat pH, uric acid, glucose monitoring, and programmed stimulation	pH (3-10); glucose (0–600 μM)	[342]
Sleep Monitor	Potential variation in the Wheatstone bridge in the circuit	Soft device with multi-layered circuit design; reduced motion artifact	Sleep quality monitoring	N/A	[307]
COVID-19 Sensor	Mechanoacoustic sensors	Soft, flexible, and miniaturized, high bandwidth, great fidelity with immunity to ambient noises, biosignal detections such as heart, respiratory rate	COVID-19 infection monitoring	N/A	[314]
"Living" Tattoo Sensor	Programmed bacterial cells as responsive components	3D printable and patternable; flexible, transparent, and gas permeable	On-skin chemicals detection with visible color changes	N/A	[185]

Abbreviation: EMF: electromotive force; **FPCBs**: flexible printed circuit boards; **EP**: electrophysiological signals; **SWH**: skin wound healing; **GSR**: galvanic skin response; **3D**: three-dimensional; **NFC**: near field communication; **HR**: heart rate; **ECG**: electrocardiogram; **EMG**: Electromyography; **EOG**: electrooculography; **EEG**: electroencephalography; **Rs**: sheet resistance; **HMI**: human machine interface; **SCG**: seismocardiogram; **SpO_2**: oxygen saturation level; **T**: temperature; **P**: pressure; **SF**: silk fibroin; **LM**: liquid metal.

[a] indicates the multifunctional skin sensors with the ability to offer information including but not limited to a combination of temperature, strain hydration, sweat pH, ECG, and EMG.

However, several challenges remain to be addressed, such as material development, electronics designs, wireless transmission/powering, data processing, energy source optimization, and integrating high-tech platform designs for customizable and personalized options.

In terms of material developments, the balance between conductivity, flexibility, skin-conformability, and biocompatibility should be deeply considered. Furthermore, cost-effective manufacturing, stable functionalities, as well as scalable production should be taken into account. With growing awareness of eco-friendliness and recyclable economics, newly recognized features, including biodegradability and self-healing capabilities, are critical from the perspective of material designs.

In addition, there are still limitations in realizing flexible wireless devices with long signaling distances and high sign-to-noise ratios, especially under dynamic strains. Developing an epidermal antenna provides a practical route to achieve high data transmission efficiency and skin-conformability. The miniaturization and integration of sensing, signaling, and power units into a single skintronic device must be further advanced to facilitate more accurate and reliable assessment and diagnosis.

As for skin-interfaced biomarker detection, despite all the extraordinary advancements in skintronics, it is essential to critically evaluate the limitation of all systems when designing next-generation sensors. Extensive studies are required to fully understand sweat biomarkers and their relationship to the human biological system. On the platform front, optimizing the biorecognition interface needs to clarify that one size approach does not fit all situations. In protein or antibody base biorecognition set-up, it offers cheap and well-understood kinetics, but it is prone to degradation and insufficient at long-term analyte monitoring. The polymer itself, though cost-effective, surface fouling and surface uniformity can be a shortcoming. Inorganics such as MOF type or carbon-based nanomaterials used as recognition interface has superb longevities, but their biocompatibility and cost remain a question.

Secondly, apart from materials at the biorecognition interface, the temporal need of the system should also be considered. If the requirement of the sensors was short-term and to be utilized in burst measurement, then the cost-effectiveness of the materials and fabrication approach should be considered. Cost-effective materials such as common polymers (PVC, polyurethanes, PMMA, and nanocellulose) based on simpler biorecognition designs using antibodies or proteins should be considered. In contrast, MXene, highly unique MOFs, or non-commercialized nanomaterials should be lowered on the materials list. However, if the system aims for long-term or continuous detection, the primary challenge will shift toward system stability, biocompatibility, and sensitivity. This would require the sensor to be constructed with more care, such as implementing protective or analyte-selective polymer layers, nanomaterials, more prolonged detection, and miniaturized electronics, as the sensor must stay on the skin without causing irritation or discomfort.

If the skintronic sensors were to enter the clinical settings, interference from other molecules would now need to be considered, as patients are likely to take other medicines that may cause false results of the sensing platform. Finally, the future of skintronic devices must expand in diversification in material selections and system constructions rather than searching for a perfect system. More tools for each situation will be more beneficial since personalized medicine requires a strategy to adapt to the patient rather than the opposite. Finally, biomarker detection from the skin still requires extensive studies to establish verification and validation parameters. Standard or commercialized sensors for biomarkers such as glucose, lactate, and chlorine have established regulatory guidelines defining product performance. However, novel biomarkers such as cancer-related small molecules have yet to be thoroughly studied. In comparison to clinical diagnostics, skin interface existing biomarkers still has a long way to go. Nevertheless, with increasing research in the field, it is not hard to see that skin interface sensing will grow into a pillar member of the next-generation medical device.

Table 2
A summary of examples for companies/startups for wearable intelligent skintronics.

Company	Headquarter Location	Product/Platform-Tech Examples	Characteristics	Application	Ref.
MC10	USA	BioStamp nPoint®	Class II FDA 510(k) cleared; soft, flexible, conformable; wireless and rechargeable; multi-modal, and multi-location sensing	Assess sleep quality	[343]
		Kintinuum™	User-comfortable; cost-effective	Quantify treatment efficacy	[344]
VivaLNK	USA	Wearable ECG sensor	Reusable, rechargeable, ultralight, and flexible; real-time data recording; IoT enabled	ECG rhythm, HR, and respiratory rate monitoring	[345]
		Wearable temperature sensor		Ambulatory and remote patient monitoring settings	[346]
Brain Scientific	USA	NeuroCap™	Pre-gelled disposable; compatible with other amplifiers of EEG signals	EEG recording in various clinic settings	[347]
LogicInk	USA	Daily Suncare Signal	Temporary and waterproof; directly applied to the skin; in control of daily sunshine exposure	Provide personal UV exposure and safe sun limit for children	[348]
		LogicInk Genetic Signal for COVID-19	Non-invasive, saliva-based colorimetric; cost-effective, at-home setting, PCR-equivalent	Early detection of COVID-19	[349]
Xsensio S.A.	Swiss	Lab-on-Skin™ Technology	Highly-miniaturized, continuous, real-time health monitoring	Provide real-time information about health and wellness	[350]
Xenoma	Japan	E-skin; sleep & Lounge and EMStyle/MEVA	Wearable, durable, washable	Provide real-time healthcare information	[351]
Company	Headquarter location	Product/platform-Tech Examples	Characteristics	Applications	Ref.
Epicore Biosystems, Inc.	USA	Gx Sweat Patch	Wearable colorimetric, visualized by smartphone App	Real-time and personalized recommendations for fitness purposes; Provide colorimetric feedback on sweat rate and sodium levels in real time	[352]
		Discovery Patch® Sweat Collection System	FDA-registered; single-used, flexible, and wearable; skin-conformable	Directly capture eccrine sweat during exercise, environmental exposure, and pharmacologic modes of sweat induction	[353]
		Connected Hydration	Being improved for worker health in heat-related incidents; wireless communication; continuous, real-time monitoring	Biomarker monitoring; provide insights and recommendations to adjust work-rest schedules	[354]

Abbreviation: ECG: Electrocardiography; **HR**: heart rate; **FDA**: Food and Drug Administration; **PCR**: Polymerase Chain Reaction.

The XR (*i.e.,* VR, AR, mixed reality (MR)) has enabled individuals to obtain various skills remotely [316,317]. Wearables allow users to experience the "real-world" scenarios created by computational algorithms, which can provide them with multiple usage options. For example, the patients can do the recovery training individually at home with remote guidance from medical professionals. However, most currently-available wearables for XR are bulky, leading to low user convenience. Hence, the skintronic platform can effectively revolutionize the usability and practicability of XR devices. Through wireless information transmission and battery-free systems, the wearers can experience real-life settings with skintronics, showing great promise in the evolution of personalized healthcare platforms. The prostheses integrated with the skintronics are empowered with the human touch, becoming a feasible solution for patients afflicted with haptic loss. Although various wearables have been developed, the eventual translation from the lab to market remains a challenge and is a critical step to prove their feasibility for clinical applications. In order to overcome these challenges, interdisciplinary research has become a significant trend for developing innovative game-changing solutions. State-of-the-art skintronic devices and their follow-up applications (summarized in Table 1) require endeavors in various fields, such as materials science, biotechnology, signal processing, and clinical medicine.

We have also witnessed many high-tech companies jumping into the research and development of skintronics for the next-generation intelligent wearables, such as MC10 and Rotex, which indicates promising commercial perspectives and opportunities in this field. Therefore, the translation from the lab bench to the market is appealing. Here we summarize current companies and the launched/approved skintronic devices in Table 2.

With the advent of the IoT philosophy and 5G, the universalization of wireless intelligent skintronics is considerably accelerating [318,319]. Furthermore, appealing concepts have been proposed to facilitate human life. For example, the Internet of Medical Things (IoMT), similar to the Internet of Healthcare Things (IoHT), envisions a network where wireless communications enable efficient healthcare data exchange between intelligent medical devices and humans [320]. Therefore, next-generation skintronics will bring a more convenient and smart lifestyle in the foreseeable future, allowing personalized and intelligent healthcare accessibility to underserved populations and underdeveloped areas.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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